Oncology, DSD, Gender & Ageing

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 24	Basic Principles of Immunology and Immunotherapy in Urologic Oncology
Ch 25	Molecular Genetics and Cancer Biology
Ch 55	Management of Abnormalities of the External Genitalia in Males
Ch 58	Management of Abnormalities of the Genitalia in Girls
Ch 59	Differences of Sexual Development
Ch 60	Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations
Ch 65	Adolescent and Transitional Urology
Ch 120	Aging and Geriatric Urology
Ch 126	Management and Considerations in Transgender and Nonbinary Individuals
Ch 127	Male-to-Female Gender-Affirming Surgery: Vaginoplasty
Ch 128	Special Urologic Considerations in Transmasculine Individuals
★	Putting it together

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

1 · Innate and adaptive immunity

Immunotherapy either switches these arms on or releases their brakes.

- P2 Describe use and principles of immunology and Immunotherapy in Urological Oncology — *innate/adaptive arms, TLR/PAMP, two signals, Th1 vs Th2, checkpoints, adaptive immune resistance, three Es*

Not in the paper answer:

- Serum IL-2 after BCG; CCR7–CCL21 homing; organ inflammation links; KEYNOTE-052 CPS rule; RCC combination trials; IMbassador250 (below).
- Bacille Calmette-Guérin (BCG) cell-wall peptidoglycans are pathogen-associated molecular patterns (PAMPs) binding **Toll-like receptor 2 (TLR2)**.
- Sentinel urothelium releases type I interferon (IFN), raising class I major histocompatibility complex (MHC), and CXCL8 (interleukin-8, IL-8), a strong neutrophil attractant.
- **The regulatory T cell (Treg) is a CD4 subset with the marker FoxP3**.
- T-helper 1 (Th1) cells (IL-2, tumour necrosis factor-α [TNF-α], IFN-γ) mean **BCG success**.
- Th2 cells (IL-4, IL-5, IL-10, IL-13) mean **BCG failure**. Raised serum IL-2 after BCG is favourable.

ALGORITHM

1. Danger signal (TNF-α, TLR ligand) reaches a dendritic cell (DC).
2. DC stops antigen uptake; raises MHC and B7-1/B7-2.
3. DC enters afferent lymphatics (CCL21 sensed by CCR7).
4. In the node it activates CD4, CD8 and B cells.

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; raised by inflammation and IFN	Antigen-presenting cells (APCs)
Peptide	~9 amino acids	13–25 amino acids

2 · Immune editing and escape

How a tumour escaped decides which drug can help.

- Escape follows equilibrium (may last a lifetime): antigen and MHC fall, Tregs rise.
- Adaptive immune resistance: CD8 cells release IFN-γ, which induces programmed death ligand-1 (PD-L1) on tumour and myeloid cells; this binds programmed death-1 (PD-1) on those CD8 cells, switching them off.

KEY POINT

PD-L1 is largely induced by the immune attack itself — a marker of an ongoing but suppressed response, not a fixed tumour trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating

T-cell molecule	Ligand	Effect
Cytotoxic T-lymphocyte antigen-4 (CTLA-4)	CD80/CD86	Inhibitory; mainly on FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly on CD8 tumour-infiltrating lymphocytes (TILs)
Lymphocyte activation gene-3 (LAG-3)	MHC class II	Inhibitory

3 · Inflammation and genitourinary (GU) tumours

Organ differences explain differing drug responses.

	Bladder	Kidney	Prostate
Does inflammation cause cancer?	Strong: *S. haematobium* (squamous), chronic urinary tract infection (UTI), catheters, cyclophosphamide	Weak (smoking, obesity)	Probable: peripheral-zone inflammation; proliferative inflammatory atrophy (PIA) beside high-grade PIN (HGPIN)
CD8 infiltration	Better outcome (muscle-invasive, MIBC)	**Worse outcome — renal paradox**	May be suppressive
Dominant restraint	PD-L1, rising with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

- Chronic Th2/Treg/M2 inflammation promotes cancer; acute Th1/CD8 response eliminates it.

4 · BCG in bladder cancer

BCG is the most successful urological immunotherapy.

ALGORITHM

1. Instil 1–5 × 10^8 bacilli.
2. A fraction adheres via **fibronectin attachment protein**.
3. Urothelial TLR2/TLR4 release IL-8 and TNF-α.
4. Neutrophil wave (hours), then DC activation.
5. CD4 Th1 response in nodes (weeks), then memory.

- Memory explains ~**50%** of non-muscle-invasive bladder cancer (NMIBC) in durable remission.
- **Nadofaragene firadenovec**: IFN-α2b protein is short-lived, so a non-replicating adenovirus carries its gene, with **Syn3** to overcome the antiadherence barrier.
- Given 3-monthly up to 12 months; carcinoma in situ (CIS) complete response (CR) 53.4%. Food and Drug Administration (FDA) 2022: **first gene therapy** for BCG-unresponsive NMIBC with CIS.

5 · Cancer vaccines

Only one vaccine has succeeded in phase III.

- P4 Rationale of using cancer vaccines in Urology — *antigen + GM-CSF adjuvant; sipuleucel-T (PAP–GM-CSF, 3 doses 2-weekly, IMPACT); failed IMA901, AGS-003, PROSTVAC*

- Sipuleucel-T prolongs overall survival (OS) by **+4.1 months** but has **no effect on time to progression**; it was the first FDA-approved solid-tumour vaccine.

6 · Checkpoint blockade

Releasing PD-1 or CTLA-4 brakes works in bladder and kidney, rarely in prostate.

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder

- First-line pembrolizumab alone (KEYNOTE-052) needs cisplatin-ineligibility with combined positive score (CPS) **≥10**, or platinum-ineligibility; with CPS <10 it did worse than carboplatin.
- Atezolizumab and durvalumab were withdrawn (failed confirmatory trials); adjuvant atezolizumab (IMvigor010) was negative.
- **Enfortumab vedotin (EV)** is an antibody–drug conjugate (anti-nectin-4 and monomethyl auristatin E). With pembrolizumab it gave an objective response rate (ORR) of 68% in cisplatin-ineligible (EV-103).
- †EV-302: EV and pembrolizumab roughly doubled progression-free survival (PFS) and OS versus platinum; now first line regardless of cisplatin eligibility.

Kidney

- CTLA-4 is on Tregs and PD-1 on CD8 TILs, so dual blockade is synergistic.
- Vascular endothelial growth factor (VEGF) impairs DC maturation and T-cell entry; a tyrosine kinase inhibitor (TKI) normalises vessels, hence TKI plus immunotherapy.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- In metastatic castration-resistant prostate cancer (mCRPC), pembrolizumab gives prostate-specific antigen (PSA) response <10%; tumours are rarely PD-L1-positive.
- Ipilimumab failed in two phase III trials (no OS benefit). Enzalutamide raises PD-L1, yet atezolizumab with enzalutamide (IMbassador250) gave no OS benefit.
- Exception: **microsatellite instability-high or mismatch repair-deficient (MSI-H/dMMR)** tumours (1.2–12%) get pembrolizumab second line or beyond. The 2017 tumour-agnostic approval was the **first based on a biomarker, not a site**.
- †Immune-related adverse events (colitis, hepatitis, pneumonitis, endocrinopathies): grade ≥2 needs drug hold and corticosteroids; endocrinopathies need lifelong replacement.

KEY POINT

Every successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, immunogenic cell death, VEGF inhibition) or removes a brake (PD-1, CTLA-4). Failures cluster where the brake is not limiting — PD-L1-negative, Treg-dominated prostate cancer.

7 · Landmark trials

These trials set where each drug sits.

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only anti-PD-1/PD-L1 approved on phase III second-line data
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk MIBC	Disease-free survival 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, metastatic renal cell carcinoma (RCC) after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS hazard ratio 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab 1 year vs placebo, high-risk clear-cell RCC	Recurrence or death reduced by 32%	Adjuvant standard

25 Molecular Genetics and Cancer Biology

1 · Hallmarks and basic molecular genetics

The vocabulary for later targeted therapy and testing.

- **Local invasion** is the one hallmark that uniquely separates malignant from benign.
- Pre-messenger RNA (pre-mRNA) undergoes 5′ capping, 3′ polyadenylation and splicing. **Alternative splicing** makes androgen receptor (AR) variants (AR-V7) lacking the ligand-binding domain; they work without ligand, causing castration resistance.
- **Somatic** testing (tumour) finds targetable mutations, resistance and heterogeneity. **Germline** testing (blood/saliva) finds hereditary syndromes for surveillance, preventive surgery and family cascade.

2 · Gene alterations: tumour suppressors and oncogenes

Oncogenes need one altered allele, suppressors usually two.

- P1 Oncogenes — *gain of function/one allele vs suppressor loss/both alleles; mutation, amplification, fusion (TMPRSS2–ERG ~half, MYC, MET); Knudson two-hit*

Not in the paper answer:

- Point mutations are silent, missense or nonsense; indels cause frameshift; rearrangements cause fusions, amplification or chromothripsis.
- Suppressor inactivation routes and two-hit exceptions (below); CDKN1B is a GU suppressor.
- Suppressors are inactivated by homozygous deletion, loss plus mutation, two mutations, or loss plus promoter methylation.
- Knudson two-hit: familial retinoblastoma is often **bilateral**, as every cell carries one hit.
- Exceptions: **dominant negative** (mutant TP53 disables wild-type in multimers) and **haploinsufficiency** (p27/CDKN1B).

3 · Cell cycle

Cancer must escape cell-cycle control.

- The cycle (G0, G1, S, G2, M) takes ~24 h and is unidirectional.
- Cyclin-dependent kinases (CDKs) are active only with their phase-specific **cyclin** (proteasome-degraded after each phase); CDK phosphorylation and CDK inhibitors (CDKIs) also regulate them.

Phase	Complex	Function
G1	Cyclin D–CDK4/6, cyclin E–CDK2	Phosphorylate RB1; cyclin D needs sustained mitogen (Ras/Raf/MEK)
S	Cyclin A–CDK2, then cyclin A–CDK1	S-phase genes
G2/M	Cyclin B–CDK1	Phosphorylates lamins, so the nuclear envelope breaks down

Checkpoint	Trigger	Mechanism
G1/S	Mitogens, nutrients, O_2, DNA damage	p21 (CDKN1A) and 14-3-3σ inactivate G1 CDKs; TP53 triggers apoptosis if damage is unrepaired

Checkpoint	Trigger	Mechanism
G2/M	DNA damage (radiation)	ATM activates CHEK1/CHEK2, which sequester Cdc25; TP53 induces p21 and GADD45
Spindle assembly	Unattached chromosome	Blocks anaphase-promoting complex; failure causes non-disjunction and **aneuploidy** (near-universal in solid tumours)

- CDKIs: **Cip/Kip** (p21, p27, p57) act broadly; **INK4** (p15, p16, p18, p19) act on CDK4/6 only, blocking RB1 phosphorylation.

RB1 — guardian of G1/S

ALGORITHM

1. Hypophosphorylated RB1 binds E2F and histone deacetylases (HDACs), so S-phase genes stay silent.
2. Cyclin D–CDK4/6 and cyclin E–CDK2 hyperphosphorylate RB1.
3. E2F is released, S-phase genes are transcribed, and the restriction point is passed (irreversible).

- RB1 loss, cyclin D amplification or CDKI loss disables it; it is altered in almost all cancers.
- **TP53** (guardian of the genome) is activated by DNA damage (via ATM/ATR and CHK1/CHK2) or by oncogene stress (ARF blocks MDM2). It then causes arrest, repair, senescence or apoptosis.
- **MDM2** is the negative regulator of TP53. TP53 is the most mutated gene in cancer (>half) and mediates both G1/S and G2/M arrest.

KEY POINT

Cancer must break both RB1 (entry) and TP53 (integrity) — so both pathways are lost in almost every tumour.

4 · DNA damage and repair

Repair defects cause cancer but are also drug targets.

- **Loeb mutator hypothesis**: early loss of replication fidelity makes a hypermutable tumour selecting resistant clones.

Pathway	Lesion	Key proteins	Cancer link
Nucleotide excision repair (NER)	Bulky adducts, cross-links (UV, chemicals)	XPC, XPD, ERCC1/XPF	Xeroderma pigmentosum; **ERCC2 in bladder cancer**
Base excision repair (BER)	Single-base damage, single-strand breaks	MUTYH, OGG1, PARP1/2	OGG1 variants linked to RCC
Mismatch repair (MMR)	Mispairs, indel loops at repeats	MSH2/MSH6, MLH1/PMS2	Mutation rate raised ~100-fold; Lynch
Homologous recombination (HR)	Double-strand break (DSB)	ATM, RAD51, BRCA1, BRCA2	Breast/ovarian, prostate
Non-homologous end joining (NHEJ)	DSB	Ku70/Ku80, DNA-PKcs, ligase IV	Translocations

- At a DSB, ATM phosphorylates TP53, CHEK1/2 and H2AX (**γH2AX** foci). Unrepaired DSBs cause apoptosis: the basis of radiotherapy and many cytotoxics.

- **HR** copies the sister chromatid (S/G2 only), so is precise. **NHEJ** needs no template and works in any phase, so is error-prone.

> **KEY POINT**
> Repair defects are targets: HR loss means PARP inhibitor sensitivity (synthetic lethality); MMR loss means immunotherapy; ERCC2 loss means cisplatin sensitivity.

5 · Apoptosis

Chemotherapy and radiotherapy kill mainly by apoptosis.

- Apoptosis is programmed, energy-requiring, with little inflammation. Initiator caspases 8, 9, 10 activate executioners 3, 6, 7.

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Unrepaired DNA damage, lost survival signals	Fas ligand/TNF on Fas (CD95)/TNFR1
Sequence	Bax/Bak release cytochrome c; Apaf-1 apoptosome activates caspase-9	FADD recruits procaspase-8, giving caspase-8
Brakes	Bcl-2, Bcl-XL, Mcl-1; IAPs	FLIP
TP53	Dependent (transactivates Bax)	**Independent**, so a target when TP53 is mutated

- Bid, cleaved by caspase-8, links the two pathways. TP53 loss causes chemotherapy and radiotherapy resistance.

6 · Senescence, telomeres, autophagy

Barriers a cell must pass to become immortal.

- P3 Telomerase — *~100 bp lost/division, two barriers, then telomerase immortalisation; normal tissue expression, ALT (ATRX/DAXX); short telomeres in premalignant lesions, predict prostate recurrence*

Not in the paper answer:

- Short telomeres also occur in premalignant bladder lesions (early, causal).
- Telomerase is high in **85–90%** of tumours.
- **Senescent** cells are viable but permanently arrested (p16, p21, p27; marker SA-β-galactosidase). Their senescence-associated secretory phenotype (SASP) is immunostimulatory but potentially pro-tumour.
- Androgen deprivation therapy (ADT) induces senescence in some prostate cancers, and SASP may explain ADT-associated inflammation.
- **Autophagy** suppresses early tumours (BECN1 loss in prostate cancer) but sustains established ones; chloroquine inhibits it.

7 · Epigenetics

Reversible changes, so biomarkers and drug targets.

- Epigenetic change alters expression without sequence change and is **reversible**.
- **DNA methylation** places 5-methylcytosine (5-mC) at CpG islands (upstream of ~40% of promoters). DNMT1 maintains methylation; DNMT3A/3B add it de novo.
- 5-mC deamination causes C-to-T transitions.
- Promoter hypermethylation silences suppressors and acts as a "second hit": **GSTP1 in >90% of prostate cancer** (commonest somatic change), RASSF1A (up to 97% of bladder cancer)

and p16.

- Global hypomethylation causes instability and re-expression of oncogenes and imprinted genes: LINE-1 (bladder) and IGF2 loss of imprinting (prostate).
- **Histones**: H3K4me3 and acetylation activate; H3K9me3 and **H3K27me3 (written by EZH2)** repress. EZH2 is raised in metastatic castration-resistant prostate cancer (mCRPC) and predicts prostate-specific antigen (PSA) recurrence.
- **Non-coding RNA (ncRNA)**: long ncRNA is >200 nt (HOTAIR drives castration resistance; MALAT1). MicroRNA (miRNA, 18–25 nt) silences **many** mRNAs (**miR-371a-3p** in testis cancer); small interfering RNA (siRNA) silences **one**.

8 · Hypoxia, VHL and angiogenesis

Explains clear cell renal cell carcinoma (RCC) and its drugs.

ALGORITHM

1. In normoxia, hypoxia-inducible factor-α (HIF-α) is prolyl-hydroxylated.
2. Von Hippel-Lindau protein (VHL, an E3 ubiquitin ligase) binds it, and the proteasome degrades it.
3. If there is hypoxia or VHL loss, then HIF-α accumulates and dimerises with HIF-1β.
4. This switches on VEGF, angiopoietin and glycolytic genes.

- The **Warburg effect** is aerobic glycolysis, best seen in clear cell RCC because HIF-1α/2α are constitutively active.
- Hypoxia also drives invasion and immunosuppression through myeloid-derived suppressor cells (MDSCs), PD-L1 and blocked dendritic-cell maturation.

KEY POINT

VHL loss = permanent pseudohypoxia; angiogenesis, Warburg metabolism, invasion and immunosuppression follow — why clear cell RCC responds to VEGF, HIF-2α and immune blockade.

- **Lineage plasticity**: under intense driver targeting, cells trans-differentiate, as in neuroendocrine prostate cancer.

9 · GU tumour genomics

Signature genes of each genitourinary (GU) cancer guide treatment.

Prostate

Gene	Significance
TMPRSS2–ERG	Commonest fusion (~half)
SPOP	Commonest mutated suppressor in primary tumours
TP53	Commonest alteration overall (more in advanced disease)
RB1, PTEN, TP53	Loss of ≥2 means aggressive variant, overlapping neuroendocrine
BRCA1/2	Germline plus somatic hit: give **PARP inhibitor**
ATM	Little PARP benefit
CDK12	Tandem duplications; poor response to hormones, chemo, PARP
MMR deficiency	Microsatellite instability (MSI): give immunotherapy

Gene	Significance
AR amplification	Late, under AR-targeted pressure; causes ADT resistance

Kidney

Subtype	Genomic hallmark	Therapy link
Clear cell (75–80%)	**3p loss** plus VHL, PBRM1, BAP1, SETD2	VEGF tyrosine kinase inhibitors (TKIs); belzutifan (HIF-2α)
Papillary (15–20%)	MET (7q); FH loss in subset	MET-active TKIs
Chromophobe	Mitochondrial DNA, mTOR pathway	mTOR-directed rational
Translocation	MiT/TFE fusions; young women	No targeted therapy
Sarcomatoid (any)	BAP1, CDKN2A loss	PD-L1-positive, so immunotherapy-responsive

- Clear cell: VHL mutated in 60–70%, methylated in 5–10%. **PBRM1** (~40%) means better outcome; **BAP1**, **SETD2** (~10% each) worse; PBRM1 and BAP1 are mutually exclusive.

Bladder (urothelial carcinoma, UC)

- UC has high mutation load (APOBEC); TP53 is most frequent, then KDM6A, ARID1A, FGFR3, PIK3CA, RB1.
- Of six consensus subtypes, **basal/squamous is largest (35%)**, luminal papillary 24%, neuroendocrine-like smallest 3%.
- **FGFR3** is altered in 15–20% of advanced UC and up to 37% of upper tract UC. They are luminal papillary and T-cell-poor (poor immunotherapy response) but respond to **erdafitinib**; test all advanced UC.
- HER2 is altered in 9–12% of muscle-invasive bladder cancer (MIBC). ERCC2 means cisplatin sensitivity (bladder-sparing trials).

Testis

- Low mutation rate; **12p gain** (†isochromosome 12p). KIT/KRAS in seminoma; TP53 mutation rare. miR-371a-3p has >90% sensitivity and specificity.

Hereditary syndromes

- P2 P1 Hereditary urological malignancies / Familial malignancies in urology — *VHL, HPRC (MET), HLRCC (FH), BHD (FLCN), BRCA2 > BRCA1, Lynch, HOXB13*

Not in the paper answer:

- VHL >70% lifetime RCC risk, Lynch prostate risk, and the germline-driver-drug link (below).
- VHL: clear cell RCC >70% lifetime; belzutifan. Hereditary leiomyomatosis and RCC (HLRCC): papillary type 2, collecting duct. Lynch: UC and prostate. HOXB13 carriers need earlier PSA screening.

> **KEY POINT**
> The germline syndrome names the somatic driver (VHL: clear cell; MET: papillary; FH: type 2; BRCA: prostate) and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss).

10 · Clinical NGS and biomarkers

Reading next-generation sequencing (NGS) reports correctly.

- **Tumour-only** panels cannot separate somatic from germline variants.
- **Variant allele frequency (VAF)** is the fraction of reads with the variant. It depends on zygosity, copy number, loss of heterozygosity and purity; a heterozygous variant at 60% purity gives 30%.
- High VAF means clonal/early; low means subclonal/late. ~50% or ~100% suggests heterozygous or homozygous **germline**; single-digit % in cell-free DNA (cfDNA) suggests **clonal haematopoiesis**.
- **Liquid biopsy** deep-sequences circulating tumour DNA (ctDNA) within cfDNA; ctDNA after treatment means minimal residual disease.
- A predictive marker is proven only by a **biomarker-treatment interaction test** in a randomised trial: parallel lines mean prognostic, crossing lines mean predictive.
- Absolute benefit scales with baseline risk: a 50% relative reduction gives 15% absolute benefit at 30% risk but 1.5% at 3%.
- **HSD3B1 (1245A>C)** raises dihydrotestosterone (DHT) synthesis, causing ADT resistance but greater abiraterone sensitivity.

11 · Landmark trials

These made genomic testing standard.

Trial	Design	Result	Take-home
†PROfound	Olaparib vs enzalutamide/abiraterone, HR-altered mCRPC	Better radiographic progression-free survival (rPFS) and overall survival, driven by BRCA1/2	Germline and somatic HR testing in metastatic prostate cancer
†THOR	Erdafitinib vs chemotherapy, FGFR-altered metastatic UC after immunotherapy	Longer overall survival	FGFR testing of all advanced UC

55 Management of Abnormalities of the External Genitalia in Males

1 · Normal development and penile size

Normal values separate a small penis from a hidden one.

- Male differentiation (weeks 9–13) needs Leydig testosterone, conversion to dihydrotestosterone (DHT) by 5α-reductase, and a functional androgen receptor (AR).
- Any break causes undervirilisation; female is the passive default.
- The genital tubercle forms the glans, urethral folds the shaft, and labioscrotal swellings the scrotum (inferomedial migration and fusion leave the raphe).
- The glanular urethra probably forms by urethral plate fusion, not ectodermal ingrowth.
- **Stretched penile length (SPL)** runs from pubis (fat pad compressed) to glans tip and correlates with erect length. At term it is **3.5 ± 0.7 cm** (diameter 1.1 cm); in adults 13.3 ± 1.6 cm.
- The micropenis cut-off is −2.5 standard deviations (SD): **1.9 cm** from term to 5 months; 1.5 cm at 30 weeks; 2.0 cm at 34 weeks; ≈ 3.7–3.9 cm at 5–11 years; 9.3 cm in adults. Compare with the cut-off, not the mean.
- Minipuberty (first 3 months): a gonadotrophin surge drives a second phase of penile growth.
- Tanner: scrotal/testicular change first (stage 2), penile length 3, girth/glans 4. Varicocele peaks at Tanner 3; semen analysis is done at Tanner 5.

2 · Prepuce: phimosis and paraphimosis

Most non-retractile foreskins are normal.

- **Physiological phimosis** is normal at birth. Smegma and erections separate it: ≈ **90% retractable by 3 years**, and < 1% remain phimotic at 17 years.
- Forceful retraction causes cicatricial (secondary) phimosis.
- Treat persistent phimosis, balanitis, posthitis, balanitis xerotica obliterans (BXO) or urinary tract infection (UTI).
- **Topical steroid**: betamethasone 0.05% twice daily for 30 days (or thrice daily for 21 days). Response comes within 2 weeks; ≈ 1 in 6 fail (more collagen, less elastic fibre).
- A retractile prepuce halves recurrent febrile UTI risk; after UTI with normal ultrasound (US), steroid for phimosis reduces recurrence.
- The commonest acquired "penile lump" is yellowish **entrapped smegma**.
- **Paraphimosis** causes oedema within hours and may progress to gangrene.

ALGORITHM

1. Compress the glans with distal traction; this succeeds in most.
2. If it fails, reduce oedema: iced glove 5 min, granulated sugar 1–2 h, or punctures.
3. If still irreducible, do a dorsal slit, then circumcision.

3 · Circumcision

Most harm comes from poor selection and technique.

- American Academy of Pediatrics (AAP) 2012: benefits are not enough to recommend routine neonatal circumcision, but they justify access and payment.
- Before using a device (Gomco, Mogen, Plastibell), separate the prepuce and inspect the meatus to exclude hypospadias and megameatus. Older children: freehand under general anaesthesia (GA).
- Analgesia is mandatory. **Dorsal penile nerve block is superior to EMLA** (eutectic mixture of local anaesthetics), whose prilocaine can cause methaemoglobinaemia; a ring block is an alternative.

Defer / avoid neonatal circumcision	Reason
Hypospadias, curvature, torsion, dorsal hood, buried or webbed penis	Prepuce needed for reconstruction
Large hydrocele / inguinal hernia	Skin is pushed forward, causing secondary phimosis or a trapped penis
Coagulopathy (contraindication); hyperbilirubinaemia (some)	Haemorrhage
Neonatal intensive care unit (NICU) setting	More complications

- **Benefits**: penile cancer occurs almost only in men not circumcised at birth. Febrile infant UTI falls (to ≥ 5 years), but **111** circumcisions prevent one UTI.
- Epididymitis and balanitis are rarer, and there is ≈ **60%** protection against HIV acquisition in men, but no reduction in transmission to partners.
- Herpes simplex virus 2 (HSV-2) and high-risk human papillomavirus (HPV) are lower. No sexual dysfunction is proven; neonatal circumcision is more cost-effective.

Complications

- P1 Circumcision undone — *0.2–5% (10–20× after infancy); bleeding commonest; adhesions/skin bridges (steroids don't help); trapped penis; revision under GA*

Not in the paper answer:

- Bleeding is more frequent in older children; occasionally a haematoma needs evacuation.
- Urethral injury needs a urethroplasty matched to the injury.
- Degloving heals by re-epithelialisation (ointment, warm baths); **no immediate suturing or graft**.
- Adhesions separate spontaneously (rare after 9 y); office lysis uses EMLA. A skin bridge can cause chordee or torsion, so excise it.
- Excess excision causes chordee or torsion (flaps, Z-plasty, split-thickness graft). Wound infection is rare (Fournier reported).
- An amputated glans should be re-attached within **8 h**.
- After penile necrosis (cautery on clamp), choose phallic reconstruction or female reassignment; the latter is problematic (androgen imprinting).

> **KEY POINT**
> Late sequelae are problems of skin quantity (too much removed: trapped penis, chordee, torsion; too little: redundancy, adhesions) or of the newly exposed meatus (stenosis).

Meatal stenosis and BXO

- P2 P1 Management of Urethral meatal stenosis / Lichen sclerosus (BXO) — *normal calibre by age, post-circumcision meatitis, upward stream, ventral meatotomy; BXO features, steroid, circumcision, buccal*

mucosa

Not in the paper answer:

- Boys with BXO present at puberty (rare < 5 years) with a non-retractile prepuce, irritation, bleeding, post-void discomfort and occasionally retention.
- Follow up those with meatal disease; buccal mucosa urethroplasty is for severe urethral disease.
- Meatal stenosis occurs almost exclusively in **circumcised** boys. Other causes are adhesion lysis, frenular devascularisation and BXO.
- Urethrography only if UTI, incontinence or proximal anomaly is suspected. Suturing mucosa to glans reduces recurrence (a sutureless clamp is equal).
- BXO in boys is rare < 5 years. Steroids help only mild preputial disease; **circumcision is preferred**, with or without meatoplasty.
- ≈ 1/5 with meatal disease need further meatal surgery. A prepuce-sparing option is triradiate preputioplasty with intralesional triamcinolone.

4 · Inconspicuous penis versus micropenis

A hidden penis needs surgery; a micropenis needs endocrine work-up.

KEY POINT

Stretched length is the single discriminator. Inconspicuous penis (buried, webbed, trapped) = normal SPL and diameter hidden by skin, fat or scar, so treatment is anatomical. Micropenis = small but normally formed, so it needs an endocrine work-up.

Buried penis type	Mechanism	Treatment
Congenital	Inelastic dartos; skin not anchored to deep fascia	Surgery
Obesity	Suprapubic fat hides shaft	Weight loss; liposuction only in adolescents
Trapped (post-circumcision)	Cicatrix embeds penis in fat	Dilation or betamethasone with retraction; then repair under GA at ≥ 6 months

- Retract skin lateral to the shaft: if the glans exposes easily, the shaft is normal. The infant fat pad resolves with walking.
- Trapped penis risks voiding difficulty, pooling with balanitis and UTI, and psychosocial harm.
- **Repair**: excise the dysgenic dartos bands (proximal dorsal) unfurl the prepuce for ventral cover, and fix dorsal subcutaneous tissue to pubic fascia and scrotal tissue to the ventral base. Skin: Z-plasty, preputial island flap or graft.
- **Webbed penis** (penoscrotal fusion) is congenital or follows excess ventral skin removal; repair by transverse incision with vertical closure, or Byars' flap. Avoid neonatal circumcision.

Micropenis

- Micropenis is normally formed but ≥ 2.5 SD below the mean (< **1.9 cm** at term), with a normal length-to-girth ratio. Testes are often small or cryptorchid.
- Differentiation is complete by week 12 under placental human chorionic gonadotrophin (hCG); a defect then gives a small penis *with hypospadias*.
- Later growth needs fetal luteinising hormone (LH), so gonadotrophin deficiency after 14 weeks gives a normally formed micropenis.
- Chromosomal causes: Klinefelter, X polysomies, trisomies 8, 13 and 18.

Cause	Examples	Hormones
Hypogonadotrophic hypogonadism (commonest)	Kallmann, Prader–Willi, Laurence–Moon–Biedl, CHARGE, hypopituitarism (hypoglycaemia, growth hormone deficiency), midline brain defects	Low LH and follicle-stimulating hormone (FSH)
Hypergonadotrophic	Rudimentary testes, gonadal dysgenesis, Robinow	High LH/FSH; no testosterone rise after hCG
Idiopathic	Delayed gonadotrophin timing	Normal axis
Partial androgen insensitivity (rare)	—	Ambiguity commoner

ALGORITHM

1. History, examination, SPL and **karyotype**.
2. Refer to a paediatric endocrinologist; measure testosterone before and after hCG, with or without a gonadotrophin-releasing hormone (GnRH) test.
3. Screen the anterior pituitary (glucose, Na/K, cortisol, thyroid); then MRI brain.
4. Test **end-organ response: intramuscular testosterone enanthate × 3 months**. Short courses spare height; prolonged ones advance bone age. Repeat at puberty.

- Most achieve male identity, standing voiding and satisfactory sexual function.

5 · Curvature (chordee without hypospadias) and torsion

Repair goes step by step from skin to corpora.

- P1 Congenital Penile curvature — *Devine–Horton types, artificial erection, plication/septotomy/graft*

Not in the paper answer:

- Curvature may be acquired after circumcision, hypospadias repair or trauma.
- A prominent frenulum causes distal chordee, treated by frenulotomy.
- Penile torsion (below) is not covered.
- Repair if sexual dysfunction is expected, with concurrent penile surgery, or if growth will worsen it.

Type	Associations	Repair (stepwise)
Ventral (commonest)	Hypospadias; dorsal hood; a prominent frenulum causes distal chordee	Deglove; excise fibrous tissue superficial to Buck's; Byars' flaps; if still curved, dorsal plication, Nesbit or corporal rotation; if the urethra is short, reconstruct it; if frenular, frenulotomy
Dorsal	Isolated, epispadias	Ventral plication; protect dorsal neurovascular bundles
Lateral	Corporal asymmetry; straight when flaccid, noticed on erection	Plicate corpus **opposite** maximal curvature

- Work superficial to deep; plicate the convex side only if curvature persists. An **artificial erection** locates the apex and confirms correction. *See Paediatric Urology book, Ch 56* for hypospadias.
- **Penile torsion** is usually **counter-clockwise (to the left)** and < 60°. It is skin-level; corpora at the base are normal.

- If < 90°, deglove, rotate the glans the opposite way and suture to skin. If ≥ 90°, mobilise the base, divide bands and place non-absorbable anchoring sutures at the corporal base, with or without a dorsal dartos flap.

6 · Abnormal number, masses and lymphoedema

Rare, but each signals associated anomalies.

Feature	Aphallia	Diphallia
Incidence / defect	1 in 10–30 million; genital tubercle fails; 46,XY, normal scrotum and testes	1 in 5 million; from accessory to complete duplication
Associations	Cryptorchidism, vesicoureteric reflux (VUR), renal agenesis, horseshoe kidney, imperforate anus, cardiopulmonary; ≈ 1/3 stillbirth/neonatal death	Hypospadias, bifid scrotum, bladder duplication, renal, pubic diastasis, anal, cardiac
Work-up / decision	Karyotype, anomaly screen, MRI; then neophallus or reassignment (reassigned may develop male identity)	Renal US, voiding cystourethrogram (VCUG), penile US/MRI; individualised

- In aphallia the urethra opens **postsphincteric** (anal verge) most often, with best survival; presphincteric (prostatorectal fistula) is intermediate; urethral atresia with vesicorectal fistula is rarest and fatal.
- Penile cysts (parameatal, median raphe, inclusion) are commonest; excise them. **Juvenile xanthogranuloma — observe 1 year**.
- Lymphoedema: Milroy (autosomal dominant [AD], at birth), Meige (1st–2nd decade) and praecox (puberty, ≈ 80%). Acquired: anogenital Crohn disease (anti-TNF treatment). Also Turner, Noonan, Klinefelter.
- Surgery is excisional (tissue between Buck's and skin), not drainage.
- Haemangiomas involute (propranolol if ulcerated). Malformations persist ("bag of worms"): image, then resect with or without embolisation. In Klippel–Trénaunay–Weber, genital lesions bleed.

7 · Urethral anomalies

Most present with abnormal stream, incontinence or UTI.

- P2 Megameatus intact prepuce / Anterior urethral valve — *MIP: wide meatus under complete prepuce, found at circumcision, pyramid repair; AUV: bulbar, diverticulum lip, VCUG, endoscopic ablation*

Not in the paper answer:

- If MIP is recognised during circumcision, stop and preserve the prepuce.
- A large anterior urethral diverticulum needs open excision with urethroplasty; post-void dribbling is a presenting feature.
- Fistula, duplication, syringocele and megalourethra (below) are not covered.
- **Congenital urethral fistula** is subcoronal with a normal meatus. Close in layers, or Thiersch–Duplay if the glans bridge is thin.
- **Urethral duplication** is usually sagittal (≈ 94%), and the **ventral channel is the functional one**.
- **Effmann** classification: type I incomplete (IA distal, IB proximal, blind); type II complete (IIA two meatuses, including IIA2 "Y-type" perineal; IIB one meatus); type III with bladder duplication.

- It presents with double stream, incontinence (worse if more proximal) and UTI. VUR is the commonest association. Evaluate with VCUG, retrograde urethrogram (RUG) and cystoscopy. **Never use the hypoplastic dorsal channel as the main urethra**.
- **Cowper syringocele** (Maizels types): simple, perforated (diverticulum-like), imperforate (obstructive cyst) and ruptured. Symptomatic ones need endoscopic unroofing.
- **Megalourethra**: **scaphoid** (commoner; spongiosum deficient, ventral ballooning) or **fusiform** (rarer; spongiosum and cavernosa deficient, flaccid penis, often lethal anomalies).
- Both go with prune-belly and upper-tract anomalies; repair by reduction urethroplasty.

8 · Priapism

In children it is mostly sickle cell disease.

- P3 Priapism: Diagnosis and Management — *ischaemic vs non-ischaemic vs stuttering, blood gas, aspiration/phenylephrine, distal then proximal shunts, embolisation*

Not in the paper answer:

- Paediatric causes, sickle-cell systemic measures, neonatal priapism and stuttering prophylaxis are in the table below.

Type (children)	Cause	Treatment
Ischaemic	**Sickle cell** (commonest), leukaemia	Hydration, O_2, alkalinisation, analgesia, with or without transfusion; then aspiration with or without phenylephrine; then shunt
Non-ischaemic	Straddle injury causing an arteriocavernous fistula	Observe; if persistent, superselective embolisation
Stuttering	Sickle cell	Bedtime pseudoephedrine; oral β-agonist; long-term phosphodiesterase-5 (PDE5) inhibitor
Neonatal	Idiopathic, birth trauma, polycythaemia	None — resolves in 2–6 days

- Corporal blood gas decides urgency. Ischaemic: dark, hypoxic, acidotic, rigid, painful (compartment emergency). Arterial: bright red, not rigid or painful.
- In sickle cell, sleep hypoventilation lowers corporal pH and O_2, causing sickling and stasis in a vicious cycle. Counsel about fibrosis and erectile dysfunction (ED).

9 · Scrotal anomalies

These often accompany hypospadias and upper-tract anomalies.

- **Penoscrotal transposition** is failed inferomedial migration; partial forms are shawl, doughnut, prepenile.
- Associations: perineal/scrotal hypospadias, caudal regression, sex chromosome anomalies and Aarskog–Scott. In the complete form, **up to 75%** have upper-tract anomalies, so do renal US and VCUG.
- Repair it with the hypospadias by **rotation-flap scrotoplasty**.

Anomaly	Feature	Association	Management
Bifid scrotum	Unfused folds	Proximal hypospadias	Scrotoplasty with hypospadias repair

Anomaly	Feature	Association	Management
Ectopic scrotum	Suprainguinal (commonest), infrainguinal, perineal	Ipsilateral renal agenesis/dysplasia, ectopic ureter; cryptorchidism; perineal lipoma usually isolated	Upper-tract US; scrotoplasty and orchidopexy at 6–12 months
Hypoplasia / agenesis	Under-developed / absent	Undescended testis (UDT); agenesis — all XY, bilateral UDT	—
Scrotoschisis	Rupture, extruded testis	Meconium peritonitis	Explore, orchidopexy, close

10 · Inguinal hernia and hydrocele

*All of these reflect where and how completely the **processus vaginalis** failed to close (≈ 18% closed at term).*

Entity	Processus vaginalis	Plan
Indirect hernia	Patent; contents enter	Repair soon after diagnosis
Communicating hydrocele	Patent, fluid only	Repair if > 1–2 y, enlarging or fluctuating
Hydrocele of cord	Isolated patent segment	Rarely resolves; repair after 1 year
Scrotal (non-communicating)	Closed	Observe — most resolve by 1 y
Abdominoscrotal	Closed; through ring (≈ 1%; bilateral 30%)	Excise/drain; ring tightening unnecessary

- Hernia affects 1–5% of children, more with prematurity (≈ 13% if < 32 weeks, ≈ 30% if < 1 kg); right to left is 3:1.
- Bilateral risk: girls, prematurity, < 1 year, cryptorchidism, ascites, ventriculoperitoneal (VP) shunt.
- An **intermittent bulge** means a patent processus, so operate; observe a fixed one. Parents' photographs and the silk-stocking sign are reliable.
- Neonatal bowel transilluminates. Use US only if obese, incarcerated or doubtful.
- Incarceration: explore immediately. Reducible: elective repair within weeks. Preterm: repair before discharge, or delay if very sick or < 1500 g.

ALGORITHM

1. Skin-crease incision; open the external oblique, sparing the ilioinguinal nerve (young infants: rings coincide, Mitchell-Banks, not opened).
2. Split the cremaster and find the sac anteromedially; separate it from the vas and vessels.
3. Perform **high ligation** at the internal ring and tighten a wide ring.
4. Finally, **confirm the testis is scrotal**.

- Peripubertal non-communicating hydrocele (> 10–12 y): scrotal approach.
- Recurrence is 0.5–1% (2% preterm, 3–6% after incarceration), mostly within 2 years. Causes: missed or torn sac, damaged floor, missed direct hernia.
- Others: iatrogenic cryptorchidism, atrophy (after incarceration), vas injury. A persistent hydrocele is observed 1 year, then repaired scrotally.
- Laparoscopic repair now matches open and sees and closes the contralateral ring.

- **Abdominoscrotal hydrocele** may cause hydroureteronephrosis and limb oedema; treat by inguinal excision with or without orchidopexy.
- Contralateral patency is ≈ 40% in infants, but a metachronous hernia develops in only ≈ **7%**. So routine contralateral exploration is not advised; laparoscopy via the sac is > 99% accurate.

11 · Acute scrotum

*Only cord torsion is time-critical. The differential includes trauma, hernia, orchitis (mumps, Henoch–Schönlein purpura [HSP]), tumour and **referred pain** (ureteric calculus, appendicitis).*

Feature	Cord torsion	Appendage torsion	Epididymitis
Frequency	20–30%	**40–60% (commonest)**	5–15%
Peak age	Perinatal; 12–16 y	7–12 y	Infancy, puberty
Onset / vomiting	Acute severe; vomiting common	Sudden or insidious; uncommon	Insidious; rare
Cremasteric reflex	Usually absent	Present	Present
Signs	High-riding, horizontal lie	**Blue dot**, upper-pole nodule	Tender epididymis; pyuria 20–40%
Colour Doppler US (CDUS)	Absent/reduced flow; **whirlpool/snail sign**	Avascular upper-pole lesion, hyperaemia	Increased epididymal flow
Management	**Emergency exploration**	Ice, non-steroidal anti-inflammatories (NSAIDs), rest	Supportive; antibiotics only if UTI

Intravaginal torsion

- The **bell-clapper** deformity is found in ≈ 12% at autopsy. Torsion is familial, commoner if cryptorchid, left-sided, ≈ 1 in 4000. Delay to presentation is the main cause of loss.
- A cremasteric reflex is present in up to 10% of torsions, so it does not exclude torsion.
- **TWIST score**: †swelling 2, hard testis 2, absent reflex 1, nausea/vomiting 1, high-riding 1. A score ≥ 5 is high risk, so go to theatre without imaging; ≤ 2 is low risk.
- On CDUS, absent flow is specific but sensitivity is only 63–90%. High-resolution cord US shows the twist in almost all. **Do not delay surgery for imaging when suspicion is high.**

Hours from onset	0–6	7–12	13–18	19–24	> 24	> 48
Orchidectomy risk	≈ 5%	20%	40%	60%	80%	90%

- Degree of twist matters (lost ≈ 540° vs 360°); transfer delay lowers salvage.

ALGORITHM

1. If torsion is suspected, explore the scrotum.
2. Detorse and apply warm packs. **Meanwhile fix the contralateral testis with non-absorbable suture**.
3. Reassess. If viable, then fix in a dartos pouch or with non-absorbable sutures. If non-viable, then do an orchidectomy; offer a prosthesis later.

- Manual detorsion fails in ≈ 1/3 (twist opposite to assumed); orchidopexy is still needed.
- Tunica albuginea incision with or without a tunica vaginalis flap for compartment syndrome is unproven.
- Late effects: atrophy despite salvage (pain > 1 day, heterogeneous echotexture), semen defects and low inhibin B. These come from ischaemia–reperfusion, not antisperm antibodies.

- **Intermittent torsion** precedes acute torsion in 30–50%, so offer **elective bilateral orchidopexy**.

Perinatal (extravaginal) torsion

- Cord and tunica twist together before scrotal fixation, giving a hard painless testis, discoloured skin or a vanished nubbin.
- Salvage is rare, and it is bilateral in 5–22%. Urgency is debated, but it is urgent if a documented normal postnatal exam came first; use an inguinal approach.

Appendage torsion, epididymitis, others

- The appendix testis (Müllerian, present in ≈ 80% of testes) and appendix epididymis (Wolffian) can twist. A blue dot may be falsely positive in cord torsion. Explore only if cord torsion is not excluded or pain is prolonged.
- Epididymitis: ask about UTI, sexual activity, catheterisation, dysfunctional voiding. Treat with ice, NSAIDs, elevation; give Gram-negative antibiotics if pyuria, and consider sexually transmitted infection (STI) treatment in adolescents.
- A prepubertal boy with a positive culture needs renal US and VCUG (ectopic ureter, urethral or ejaculatory duct anomaly); recurrent cases need endoscopy.
- Others: idiopathic scrotal oedema (eosinophilia, **fountain sign**; antihistamine), HSP (may cause torsion) and cystic dysplasia of the rete testis (ipsilateral renal agenesis).

12 · Adolescent varicocele

Most adolescent varicoceles need no surgery.

- P1 P3 Varicocele – Pathophysiology / Varicocele: controversies — *left-sided anatomy, nutcracker, grading, right/non-decompressing: image for mass; adolescent: observe, repair for asymmetry*

Not in the paper answer:

- Prader or Takihara orchidometers are acceptable, though US is most sensitive for asymmetry.
- Semen analysis is the best indicator once growth stabilises.
- For microscopic repair, a microscope is better than loupes.
- Prevalence is ≈ **15%**; it appears after 10 years, peaks at Tanner 3, favours tall thin boys and is familial.
- Measure testicular volume by **US (L × W × H × 0.71)**. Volume doubles from Tanner 2 to 3, so serial measures are needed. Peak retrograde flow > 38 cm/s correlates with asymmetry.
- Surrogates do not reliably predict adult semen, and catch-up growth may be spontaneous, so **observation is the default**.
- An intratesticular varicocele (veins > 2 mm) with asymmetry should be repaired.

> **KEY POINT**
> **Indications for repair**: left testis > **20%** smaller (or bilateral hypotrophy), pain, abnormal semen (Tanner 5 / ≥ 18 y), intratesticular varicocele with asymmetry. Total volume < 30 mL and low total motile count predict poor semen.

Technique	Recurrence	Hydrocele	Note
Open retroperitoneal (Palomo, mass ligation)	2–4%	Up to 30%	Lymphatics ligated
Laparoscopic, non-sparing	0–9%	11–32%	—

Technique	Recurrence	Hydrocele	Note
Laparoscopic artery/lymphatic-sparing	1–7%	0–4%	Lymphatic sparing cuts hydrocele
Microscopic subinguinal	0–10%	0–6%	Best balance; rare atrophy
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy (retrograde/antegrade)	6–35% / 4–12%	Occasional	Mainly for recurrence

- At laparoscopy, clipping veins accompanying the artery prevents recurrence. Atrophy risk comes from artery ligation followed by later surgery on the vasal vessels.
- Hydroceles may appear ≥ 2 years later, and up to half need intervention. Genitofemoral nerve injury causes thigh paraesthesia.
- Sclerotherapy is retrograde transfemoral (technical failure 5–22%; radiation) or antegrade scrotal, using sodium tetradecyl sulfate, polidocanol or coils.

13 · Epididymal and vasal anomalies

Absent vas points to a cystic fibrosis gene or kidney anomaly.

- Epididymal cysts are common. Spermatoceles are postpubertal and contain sperm; most are observed.
- Bilateral absent vas suggests **CFTR** mutation; unilateral absence suggests **renal anomalies** (Wolffian origin).
- Persistent mesonephric duct: vasoureteral fusion with ipsilateral renal and seminal vesicle agenesis, causing UTI or epididymo-orchitis.
- For microlithiasis and undescended testis, *see Paediatric Urology book, Ch 57*.

58 Management of Abnormalities of the Genitalia in Girls

1 · Embryology that explains the anomalies

Why this matters: embryology predicts the associated anomalies.

- The **urorectal septum** divides the cloaca into urogenital sinus (ventral) and anorectal canal. Its node, the **perineal body**, splits the cloacal sphincter into urogenital diaphragm and external anal sphincter, so **both are pudendal-innervated**.
- Müllerian ducts arise lateral to wolffian ducts, so a **müllerian anomaly travels with an ipsilateral renal anomaly**. Fused ducts form uterus, cervix and upper two-thirds of vagina; **sinovaginal bulbs** form the distal vagina.
- Remnants: Skene (prostate), Bartholin (bulbourethral), Gartner duct (wolffian).
- **Caudal regression**: weeks 4–5 insult, linked to maternal diabetes; earlier insult gives higher anorectal malformation. Cranial extension gives VATER or MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).

2 · Evaluation and classification

Why this matters: whether it is obstructive sets the timing of surgery.

- Examine frog-leg; the **pull-down manoeuvre** (labia majora drawn caudally and laterally) shows the distal vagina. Discharge or bleeding needs vaginoscopy under anaesthesia.
- Ultrasonography (US) shows anatomy and upper tracts; **magnetic resonance imaging (MRI) is the gold standard** for müllerian anatomy. The American Society for Reproductive Medicine classification is broadest.

Category	Mechanism	Examples
Hypoplasia/agenesis	Fails to form	Mayer–Rokitansky–Küster–Hauser (MRKH)
Vertical fusion	Faulty müllerian–sinus canalisation	Imperforate hymen, transverse septum, distal atresia
Lateral fusion	Failed midline fusion/resorption	Didelphys, Herlyn–Werner–Wunderlich syndrome (HWWS)

KEY POINT

Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery.

3 · Clitoris

Why this matters: newborn clitoromegaly may mean salt-wasting congenital adrenal hyperplasia (CAH).

- **Clitoromegaly** is commonest from **CAH**, so check electrolytes, 17-hydroxyprogesterone and karyotype *(Ch 59)*. Others: maternal androgen tumour, Turner mosaic disorder of sex development (DSD), plexiform neurofibroma.
- Normal corpora with a large hood is **pseudo-enlargement**. A **diminutive clitoris** suggests complete androgen insensitivity syndrome (CAIS).

4 · Vestibule: the interlabial mass

Why this matters: the meatus position usually gives the diagnosis.

- **Urethral prolapse** affects prepubertal African-ancestry girls and postmenopausal women. A **doughnut with central meatus** needs no imaging. Sitz baths and topical steroid or oestrogen come first (recurrence high), then **circumferential excision**; avoid ligation over a catheter and cryosurgery.
- A **urethral polyp** is benign; excise it. A **Skene cyst** (maternal oestrogen) gives a **displaced meatus, eccentric stream**; it usually ruptures.
- A **Gartner cyst / covered ectopic ureter** sits on the anteromedial vaginal wall; incise it. Upper-pole heminephrectomy if wetting or infection persists.
- A **prolapsed ureterocele** (duplex upper pole) gives an infant retention and a pink-to-dusky mass. Decompress, then reduce, usually under general anaesthesia.
- An **ectopic ureter** below the sphincter causes **daytime wetting despite normal voiding** (†continuous dribbling). Single-system ectopia may mimic an absent kidney.

> **KEY POINT**
> Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Any cyst of urethral or vaginal origin needs a **renal US, always**.

- A **labial swelling** is commonest an inguinal hernia. **Childhood asymmetric labium majus enlargement (CALME)** is physiological: observe. If it grows, exclude neurofibroma, sarcoma, vascular malformation, Crohn disease.

5 · Vagina: obstruction and absence

Why this matters: block level and uterus presence decide the operation.

Entity	Level / uterus	Clue	Treatment
Imperforate hymen	Hymen; normal	Whitish bulge (neonate); bluish bulge and amenorrhoea (adolescent)	**Transverse** incision; not aspiration
Transverse septum	Upper/middle third; normal	Normal hymen, blind vagina	Excision, Z-plasty, balloon; hollow mould
Distal atresia	Sinus; normal	Shallow dimple; distended vagina per rectum	Pull-through with or without flaps
MRKH	Bulbs/plate; absent	Hymenal fringe and pouch; 46,XX	Dilation first
Cervical agenesis	Cervix; uterus present	Obstruction; MRI	Often hysterectomy and neovagina
HWWS	Lateral fusion; didelphys	Menstruates; lateral bluish bulge	Wide septal incision

- **Imperforate hymen** is probably the commonest obstructive anomaly. Transverse incision avoids urethra and rectum; aspiration causes ascending infection. Hymenal injury suggests abuse.
- **Transverse septum** occurs in ~1 in 70,000. Delay causes retrograde menstruation and so endometriosis. **MRI distinguishes a high septum from cervical agenesis**.
- A gonadotropin-releasing hormone agonist or continuous pill lets the lower segment dilate. **Z-plasty** turns a transverse scar longitudinal. The main complication is **stenosis**. Use a **hollow**

mould when a uterus is present.

- **Distal atresia**: the haematocolpos acts as tissue expander; drain, then pull through.

MRKH (müllerian agenesis)

- MRKH occurs in ~1 in 5000: 46,XX with **functioning ovaries**. It is the second commonest primary amenorrhoea, after gonadal dysgenesis. A functioning rudimentary uterus causes cyclic pain.
- Renal anomaly occurs in ≈ **one-third**, skeletal in 10–20%; no lumbosacral dysraphism. **Type A**: symmetrical buds, normal tubes, no extra-müllerian anomaly. **Type B**: asymmetrical with renal, Klippel–Feil or MURCS anomalies; needs full renal and skeletal work-up.

CAIS: *Ch 59*; urogenital sinus and cloaca: *Ch 60*.

6 · Vaginal replacement

Why this matters: dilation alone usually works, so surgery is for failures.

- **Dilation is first-line** (Frank; Ingram bicycle seat). It needs a dimple and compliance; surgery is no better. **Vecchietti** uses an olive with laparoscopic traction.
- **McIndoe** puts a split-thickness graft over a stent in the urethrorectal space; it stenoses, so dilation is mandatory. Full-thickness graft or Williams labial pouch contracts less. Alternatives: **buccal mucosa**, **robotic peritoneal vaginoplasty** (no bowel, no hair).
- **Sigmoid** is the default bowel. Excise ~3 cm of intervening bowel so suture lines do not overlap (fewer fistulae); fix to posterior peritoneum against prolapse. **Ileum** stenoses more; use it only for an irradiated pelvis or absent colon.
- Bowel lubricates and contracts less but risks discharge and **adenocarcinoma**; skin risks stenosis, **squamous carcinoma**, condylomata. Every neovagina needs **annual examination**.

> **KEY POINT**
> **Cervical atresia**: a perineum-to-uterus anastomosis loses the endocervical barrier, causing life-threatening ascending infection. Hysterectomy with neovagina is often best; otherwise endometrial suppression and in vitro fertilisation with transmyometrial transfer.

7 · Lateral fusion anomalies

Why this matters: HWWS girls menstruate, so obstruction is missed.

- **Didelphys** has a septate vagina in up to 75%; excise it only for dyspareunia, obstructed delivery or two tampons. Incise a **septate uterus** only for recurrent pregnancy loss.
- **HWWS** is didelphys with a unilateral imperforate hemivagina: pain **without amenorrhoea**, a lateral bluish bulge and **ipsilateral renal agenesis**. Treat by wide septal incision. Zhu class 1 is complete (earlier); class 2 is perforated (commoner, later).

8 · Acquired disorders

Why this matters: do not miss rhabdomyosarcoma or abuse.

- **Vaginal rhabdomyosarcoma**: grapelike bleeding mass under 2 years; best prognosis of genital primaries (embryonal). Biopsy, then computed tomography (CT) and marrow, then chemotherapy with or without resection. Intergroup Rhabdomyosarcoma Study (IRS) trials cut surgery from **100% to 13%**.

- **Labial adhesions**: commonest interlabial abnormality, peak 13–23 months; minora fuse (majora fusion suggests DSD). **80% resolve in 1 year**, so treat only symptoms.
- Use topical oestrogen (breast budding, pigmentation) or betamethasone 0.05% with pull-down; next, office separation under EMLA (anaesthetic cream); last, surgery. Recurrence depends on **family compliance**.
- P1 Female circumcision — *FGM ~200 million, WHO types I–IV (III = infibulation), complications, midline deinfibulation*
- **Condylomata** (human papillomavirus, HPV) suggest abuse, but perinatal transmission occurs. Over 50% resolve; use imiquimod or CO_2 laser; vaccination prevents.
- An **inguinal hernia** may present CAIS (~1–2.5%). Look for a uterus on US or a cervix, or check karyotype; the **round ligament is absent in genetic males**. If CAIS, preserve testes (aromatised oestrogen).

59 Differences of Sexual Development

1 · Definition and nomenclature

Why this matters: modern terms are karyotype-based, so the name starts the diagnosis.

- In a **difference of sex development (DSD)** chromosomal, gonadal or anatomical sex is atypical. The **2006 Chicago consensus** (Lee) replaced "intersex", "hermaphrodite" and "sex reversal" with karyotype-based terms; "atypical" is preferred to "ambiguous".

Old term	2006 term
Male pseudohermaphrodite	46,XY DSD
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
XX male / XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis (not androgen insensitivity syndrome, AIS) MCQ

- Offer endocrine and genetic testing for **proximal hypospadias even with descended testes**.

2 · Normal sex development

Why this matters: any DSD follows from which hormone was missing, where and when.

- Chromosomal sex sets gonadal sex (bipotential until ≈ 6 weeks), which sets phenotypic sex (bipotential until ≈ 8 weeks).
- **SRY** (sex-determining region Y) upregulates **SOX9**, making Sertoli cells. Genes acting elsewhere explain syndromic DSD: **WT1** (Wilms tumour, nephropathy), **SF1** and **DAX1** (adrenal failure), SOX9 (campomelic dysplasia). DAX1 duplication (Xp21) causes 46,XY dysgenesis.
- The ovary needs RSPO1/WNT4/β-catenin and FOXL2.
- **Sertoli anti-Müllerian hormone (AMH)** from 7–8 weeks makes Müllerian ducts regress by ≈ 10 weeks.
- **Leydig testosterone** from ≈ 9 weeks (autonomous, then placental human chorionic gonadotropin (hCG), then fetal luteinising hormone (LH)) builds the Wolffian ducts.
- **5α-reductase 2 makes dihydrotestosterone (DHT)**, which binds the receptor more strongly; it is the active androgen for prostate, urogenital sinus and external genitalia. Virilisation is complete by 12–13 weeks; penile growth and descent follow in the third trimester (androgen, INSL3).

> **KEY POINT**
> The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. A uterus in 46,XY means Sertoli (AMH) failure — dysgenesis, not a synthesis or receptor defect.

3 · Classification (2006 / Chicago)

Why this matters: karyotype sorts every case into three groups, each with a short cause list.

Category	Conditions
Sex chromosome DSD	47,XXY Klinefelter; 45,X Turner; 45,X/46,XY mixed gonadal dysgenesis (MGD); 46,XX/46,XY chimerism
46,XX DSD — gonadal	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (SRY+, SOX9 duplication)
46,XX — androgen excess	Fetal: congenital adrenal hyperplasia (CAH: 21-hydroxylase, 11β-hydroxylase, 3β-HSD, P450 oxidoreductase); fetoplacental: aromatase deficiency; maternal: luteoma, drugs
46,XX — other	Cloacal anomalies; Mayer–Rokitansky–Küster–Hauser (MRKH); vaginal atresia
46,XY DSD — gonadal	Complete/partial gonadal dysgenesis (SRY, SOX9, SF1, WT1, DHH, MAP3K1); ovotesticular; testis regression
46,XY — androgen synthesis	LH receptor; Smith–Lemli–Opitz; StAR; 3β-HSD; 17α-hydroxylase; 17β-HSD3; 5α-reductase 2
46,XY — androgen action	Complete and partial AIS (CAIS, PAIS)
46,XY — other	Persistent Müllerian duct syndrome (PMDS); vanishing testis; isolated hypospadias; hypogonadotropic hypogonadism; cryptorchidism

- 3β-hydroxysteroid dehydrogenase (3β-HSD) and P450 oxidoreductase deficiency virilise 46,XX yet undervirilise 46,XY. 21-hydroxylase deficiency (21-OHD) leaves 46,XY genitalia normal.

4 · 46,XX DSD

Why this matters: CAH is commonest and can kill a newborn through salt loss.

CAH — 21-hydroxylase deficiency

- CAH is the commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-OHD is **95%** of CAH; autosomal recessive, ≈ 1 in 15,000.
- Low cortisol raises adrenocorticotropic hormone (ACTH), driving androgen excess; low aldosterone causes **salt wasting**.
- 46,XX infants have normal ovaries and Müllerian structures, often a **common urogenital sinus**; some look like a boy with **bilateral impalpable gonads**. Virilisation is graded **Prader I–V** (†V = male-appearing phallus, meatus at tip, fused empty scrotum).
- **17-hydroxyprogesterone (17-OHP) is raised**; measure after **36 h**. Androstenedione, testosterone and renin rise and cortisol falls; a cosyntropin test helps.
- A **salt-wasting crisis** (low sodium, high potassium, acidosis, shock) usually comes no earlier than **day 5–7**.
- Treat with glucocorticoid, mineralocorticoid and salt (†hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone, NaCl in infancy; double-to-triple in illness or surgery).
- Affected boys look normal, so **newborn screening** matters most.
- Long term, balance androgen suppression against steroid excess; "backdoor" androgens hinder control; fertility is reduced.
- Males may get **testicular adrenal rest tumours**: benign and regress with more glucocorticoid, but obstruct tubules, so bank sperm; excision is rare. Poor suppression causes hypogonadotropic hypogonadism.
- Prenatal dexamethasone (by 6–7 weeks; ≥ 4 fetuses treated per girl helped) is experimental only.

- **Classic salt-wasting** (crisis in first weeks); **simple virilising** (no shock); **nonclassic** (20–50% enzyme activity; premature adrenarche, hirsutism, oligomenorrhoea; may need no treatment).

Enzyme	Share	BP / K^+	46,XY	Signature
21-hydroxylase	95%	Salt wasting; low BP, high K^+	Normal	High 17-OHP
11β-hydroxylase	≈ 5%	Hypertension, low K^+	Normal	—
3β-HSD	<1%	Salt wasting	Undervirilised	High dehydroepiandrosterone (DHEA)
17α-hydroxylase	Rare	Hypertension, low K^+	Female/atypical; delayed puberty	Low androgens

Other 46,XX DSD

- P1 P2 46-XX Male — *SRY translocated to X; testes, no Müllerian structures; azoospermia (no AZF)*
- Aromatase (CYP19) deficiency virilises mother and infant and recurs in later pregnancies. Maternal androgen or aromatase exposure ends at birth, so no postnatal therapy is needed.

5 · 46,XY DSD

Why this matters: a uterus separates dysgenesis (no AMH) from synthesis or receptor defects.

Condition	Key features
Complete gonadal dysgenesis (Swyer)	Streaks, female genitalia, uterus; absent puberty; tumour 30–45%, so prophylactic gonadectomy; oestrogen; donor-ovum pregnancy
Partial gonadal dysgenesis	Dysgenetic testes, atypical genitalia, variable Müllerian structures. WT1: Denys–Drash (congenital nephrotic syndrome, Wilms 90%) or Frasier (late focal segmental glomerulosclerosis, gonadoblastoma 60%, no Wilms)
CAIS	X-linked; female genitalia, short blind vagina, no uterus; breasts by aromatisation; ≈ 50% have inguinal hernia; 1–2% of girls with hernia are CAIS
PAIS	Atypical genitalia; identity mostly male; pubertal breast growth
5α-reductase 2	Autosomal recessive; atypical/female at birth, blind pouch; **virilises at puberty**, fertility reported; testosterone:DHT **>20:1**; "guevedoces" (Imperato-McGinley 1979)
17β-HSD3	High androstenedione:testosterone ratio; virilises at puberty
LH receptor defect	Leydig hypoplasia; no hCG response
PMDS	AMH or AMH-receptor mutation (autosomal recessive); male genitalia with uterus and tubes; found at hernia repair or orchidopexy; 30–50% of transverse testicular ectopia
Vanishing testes	Loss after first trimester: male genitalia, impalpable testes, no Müllerian structures; undetectable testosterone/AMH, high gonadotropins at mini-puberty; testosterone from adolescence

- In synthesis and action defects AMH is normal, so no uterus; adrenal failure occurs only if the enzyme is shared with cortisol synthesis.
- PAIS raised male may respond to high-dose testosterone; raised female, use a gonadotropin-releasing hormone agonist or gonadectomy, then oestrogen.
- In CAIS dilation usually suffices. At herniorrhaphy in a girl, seek a cervix (vaginoscopy) or a testis in the sac.

- **PMDS (Clarnette)**: 1, bilateral intra-abdominal (60–70%); 2, testis in sac or scrotum with contralateral hernia (20–30%, "hernia uteri inguinale"); 3, transverse ectopia (10%).
- PMDS needs orchidopexy. The vasa hug the uterus, so remnants were kept; Müllerian malignancy (3–8%) means weighing removal (laparoscopic, sparing vasa) against imaging.

6 · Sex chromosome DSD and ovotesticular DSD

Why this matters: Y-linked tumour risk and Turner heart and kidney disease.

- P4 P3 Klinefelter's syndrome / Mixed gonadal dysgenesis — *47,XXY features, hormones, micro-TESE; MGD 45,X/46,XY asymmetry, second commonest atypical genitalia, tumour 15–35%, remove streaks, Turner screening*

Not in the paper answer:

- Klinefelter: check gonadotropins and testosterone at mini-puberty; no testosterone in infancy.
- MGD: the impalpable side has a flat labioscrotal fold. Raised female, the testis may be relocated to groin or labioscrotal fold for surveillance.
- **Turner with Y** material: tumour 12–20% (gonadoblastoma, then dysgerminoma). The 2017 guideline advises gonadectomy at diagnosis, on low-level evidence now challenged.
- Klinefelter affects 1 in 600 males.
- **Turner** (25–50 per 100,000): short, absent puberty, webbed neck, coarctation, bicuspid valve, horseshoe kidney (renal anomaly ⅓–½); ovarian insufficiency ≈ 95%. Give lifelong cardiac and renal screening, growth hormone, oestrogen and progesterone, and oocyte donation or cryopreservation.
- In MGD, ≈ 85% of 45,X/46,XY have male-typical genitalia; dysphoria affects a substantial minority raised female.
- **Ovotesticular DSD** has ovarian and testicular tissue; most are 46,XX, with a sinus and uterus. A **vas always accompanies a testis**, a tube an ovary.
- The ovary is usually normal, the testis dysgenetic; tumour ≈ 3%. Biopsy both poles of a bipolar ovotestis. Defer irreversible steps; puberty is usually oestrogenic.

7 · Evaluation of the newborn with atypical genitalia

Why this matters: first exclude salt-wasting CAH; designation can wait for the team.

- P3 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — *history, exam, gonads as key sign, karyotype/FISH, 17-OHP, 1-week hormone table, karyotype-based algorithm, sex of rearing, surgery timing, gonadal risk*

Not in the paper answer:

- 46,XX without two ovaries suggests ovotesticular DSD; 46,XY with a uterus, partial dysgenesis or ovotesticular DSD.
- No testosterone rise after hCG may also mean no testis. High androstenedione also suggests 17β-HSD3.
- Late presenters: CAIS, Swyer, MRKH, Turner with Y, Klinefelter, 5α-reductase. After puberty, high LH and testosterone with female genitalia means CAIS.
- Examine adolescents with few clinicians, only when management may change.

> **KEY POINT**
> **Emergency**: atypical genitalia with bilateral impalpable gonads = salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP; watch for crisis from the end of week 1. Designation and surgery are not emergencies.

- A multidisciplinary team (MDT) reviews within **24–48 h**. Say "your baby", never "it". A urologist without a team transfers early.
- Count palpable gonads (ovaries do not descend): none suggests CAH; one, MGD, ovotesticular DSD, PAIS or partial dysgenesis; two, 46,XY DSD.
- Hypospadias with undescended testis (Kaefer 1999): DSD ≈ 30%; 50% if testis impalpable; 65% if meatus proximal.

ALGORITHM

1. Send karyotype (≥45 cells, for mosaicism; fluorescence in situ hybridisation for X/Y in hours), electrolytes, and 17-OHP at ≈ 36 h.
2. Pelvic ultrasonography for uterus and gonads.
3. At ≈ 1 week: LH, follicle-stimulating hormone (FSH), testosterone, oestrogen, androstenedione, DHT, AMH, with or without hCG stimulation.
4. Genetic panel, microarray or exome.
5. If still unclear, then magnetic resonance imaging, genitography, endoscopy, laparoscopy with or without biopsy (**no gonadectomy during diagnosis**).

- Normal AMH means functioning Sertoli (or granulosa) cells; high LH and FSH, gonadal failure; DHEA sulfate rises in 21-OHD and 3β-HSD.
- A precise diagnosis is nearly always made in 46,XX DSD but in **<50%** of 46,XY.
- After puberty, high gonadotropins with low sex steroid mean gonadal failure.

8 · Gender assignment and MDT

Why this matters: a wrong sex of rearing is hard to undo.

- The team adds psychologist, ethicist and peer support in a **same-day joint clinic**; care is lifelong, with transition.
- The **Money (1975)** model (early designation, early surgery, secrecy) failed; transparency and shared decisions replaced it.
- Sex of rearing predicts satisfaction better than androgen exposure does; no test predicts identity.
- Female trend: 46,XX CAH, CAIS, Swyer, LH receptor defect. Male trend: PAIS, 5α-reductase, 17β-HSD3, partial dysgenesis.
- Dysphoria is highest in 5α-reductase deficiency raised female (≈ half), lowest in 46,XX CAH raised female.
- Micropenis (<2.5 SD; <2 cm at 0–5 months) needs an endocrine work-up; female reassignment is no longer recommended.

9 · Surgery

Why this matters: genital surgery is often irreversible, so timing and consent matter.

- **Feminising** surgery (clitoroplasty, vaginoplasty or sinus mobilisation, labioplasty; *Ch 60*): clitoral surgery is irreversible; infant benefit is unproven. Only retention or obstruction is urgent.
- **Masculinising** surgery (proximal hypospadias with or without penoscrotal transposition, orchidopexy) has high complications but is revisable, so infant repair is often chosen.
- Ideally defer irreversible surgery until the individual can take part; CAH is debated.

- In MRKH (renal anomaly ≈ ⅓, so renal ultrasound), dilation comes first. Uterine remnants are removed or suppressed, **never joined to a neovagina** (no cervix). Uterus transplantation is investigational.

10 · Gonadal tumour risk

Why this matters: some gonads must come out; others are kept for hormones.

Risk rises with Y material (gonadoblastoma locus, TSPY), dysgenesis and intra-abdominal position, and falls with a scrotal gonad. Practice now weighs individual risk, function and autonomy.

Risk	Condition	Germ cell tumour	Management
High	Dysgenesis with Y, intra-abdominal (Swyer, MGD)	15–35% (Swyer 30–45%)	Gonadectomy at diagnosis
	Frasier	60%	Gonadectomy
	Denys–Drash	40%	Gonadectomy
	PAIS, non-scrotal	50%	Gonadectomy
Intermediate	Turner + Y	12%	Gonadectomy (debated)
	17β-HSD3	28%	Monitor or gonadectomy
	PAIS / dysgenesis, scrotal	Unknown	Biopsy at puberty (≥30 tubules, OCT3/4)
Low	CAIS	2%	Keep until after breast development
	Ovotesticular DSD	3%	Remove discordant testicular tissue
?None	5α-reductase; Leydig hypoplasia	Unknown	Unresolved

11 · Ethics

Why this matters: these principles govern every decision.

- Respect autonomy, be transparent, and avoid irreversible procedures before consent unless necessary. Parental consent covers limits of predicting identity and fertility.
- Psychosocial morbidity is higher, so psychology is integral. Respect preferred terms.

60 Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

1 · Principles

Why this matters: the debate is when to operate and who decides.

- P3 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — *classification, work-up, CAH care, team, sex of rearing, gonadal risk*

Not in the paper answer:

- Two descended gonads make 46,XX very unlikely; hypertension suggests 11β-hydroxylase deficiency; Huffman index >10 mm² is a concern.
- Biopsy gonads only if the sex of rearing would change; a cervical impression on genitogram means female organs.
- Endocrine Society 2018: families with experienced surgeons decide. Never use a flap for a high confluence. Introitoplasty after puberty in 25–100%.
- The controversy is **timing and who decides**, not technique. Present pros and cons of surgery *and* no surgery; never persuade.
- Support whichever path is chosen and offer all support groups. Diagnosis and gender assignment: *Ch 59*.

2 · Urogenital sinus and cloaca — classification

Why this matters: true urethral length decides which operation is safe.

In a **urogenital sinus** the vagina still communicates with the urinary tract (usually the mid–distal urethra), and both exit as one **common channel**.

Entity	Distinguishing feature
Genital atypia states	Commonest; mostly congenital adrenal hyperplasia (CAH)
Pure urogenital sinus	Normal external genitalia; behaves like a cloaca
Persistent cloaca	Rectum also joins, so there is **one perineal opening**
Female exstrophy	Probably a form of urogenital sinus

- CAH causes half of all newborn atypical genitalia and **90%** of 46,XX cases. Salt wasting occurs in **67–75%**; get a karyotype within **48 h**.
- Cloaca occurs in ≈ 1 in 20,000–50,000 and makes up 5–13.6% of anorectal malformations.

> **KEY POINT**
> **Governing principle**: **true urethral length** (from bladder neck to confluence) decides the operation far more than common-channel length, in sinus and cloaca. Confluence level is a continuum, not just "high" or "low".

- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants). Virilisation lengthens the **vagina-to-meatus** distance, so a "high" CAH confluence is an illusion.
- The **Rink classification** records common-channel length, bladder-neck-to-vagina distance, confluence level, clitoral size and appearance.

3 · Clinical evaluation

Why this matters: the first newborn dangers are metabolic and obstructive.

- The **first threat is metabolic**: salt wasting causes adrenal crisis, so monitor electrolytes before reconstruction. **Hypertension** suggests 11β-hydroxylase deficiency.
- A **suprapubic mass** (bladder, **hydrometrocolpos**, or both) is often the only early sign of a pure sinus.
- The **Huffman index** (glans width × length) is normally **<3.5 mm²**; **>10 mm²** is a concern. Isolated clitoral hypertrophy in a preterm is not a DSD.
- If **both gonads are descended, 46,XX is extremely unlikely**. CAH causes labioscrotal hyperpigmentation.
- In a **posterior cloaca** the sinus opens into the anus (type A) or just in front of it (type B).

Hydrocolpos in cloaca forms because urine refluxes into poorly draining vagina(s), adding to mucus stimulated by maternal oestrogen. It is commoner with a common channel **>3 cm**, duplication and posterior cloaca. Severe distension causes respiratory distress, cyanotic legs and acidosis.

Associated (cloaca)	Findings	Frequency
Renal	Solitary kidney, dysplasia, pelviureteric junction obstruction, ectopia	Majority
Uterovaginal	Duplication/septation	**53–67%**
Spinal	Sacral anomalies; tethered cord ≈ one-third (mostly no skin stigmata)	Cord anomaly on magnetic resonance imaging (MRI) ≤ 43%
VACTERL; gut	Tracheo-oesophageal fistula; malrotation, duodenal atresia	Correct malrotation at colostomy

> **KEY POINT**
> Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). Number of perineal orifices tells you which world you are in.

4 · Imaging and endoscopy

Why this matters: CAH needs little imaging; a cloaca needs a full 3-D map.

Modality	CAH sinus	Pure sinus / cloaca
Ultrasonography (US)	Mandatory; **cerebriform adrenals**	Mandatory for kidneys; poor for Müllerian anatomy
Genitography	Least useful; often inaccurate	**Mandatory**, as a computed tomography (CT) or MRI cloacogram
MRI	Limited	Pelvis, spine, sphincter muscles
Endoscopy	At reconstruction	Early, separate: decompress and define

- On a **genitogram**, a **cervical impression** at the vaginal dome means female internal organs.
- Do a **gonadal biopsy** only if it would change the gender of rearing. Use a **deep longitudinal incision**, because ovarian tissue may surround the testicular part of an ovotestis.
- **Prenatal cloaca** (from 19 weeks) shows fetal ascites, bilobed pelvic cysts, hydronephrosis and reduced liquor. **Always do echocardiography.**

- For a **cloacogram**, catheterise bladder, vagina(s) and rectum, then reconstruct in 3-D on CT or MRI. It shows confluences, urethral length, vaginas and the true rectum (may lie well above the fistula).
- Duplicated vaginas (≈ half) lie side by side; the rectal fistula most often opens **within the vaginal septum**.
- At **early endoscopy** a distended vagina compresses the bladder, so **empty the vagina first**.

Common channel (Peña)	<3 cm	>3 cm
Anomalies; hydrocolpos	Fewer	More (nearly all have urological defects)
Abdominal approach	Rarely	More likely
Bladder	Better spontaneous voiding	Clean intermittent catheterisation (CIC) far more often

5 · DSD reconstruction — timing and consent

Why this matters: no trial compares early with late surgery.

- **For early surgery** (historically at 2–6 months): tissue is pliable, trauma is less and parents feel relief.
- **Against early surgery**: advocacy and human-rights bodies seek a moratorium until the child can consent, because gender identity is unknown.
- Most CAH patients and parents support parental decision. All families see a psychologist; observation is always offered.
- **Clitoroplasty** is only for severe hypertrophy (**Prader III–IV**); it is the most controversial component.

High-confluence vaginoplasty	Arguments
Early single-stage	Redundant clitoral/sinus tissue is usable; maternal oestrogen thickens tissue; a planned one-stage repair does better
After puberty	Infant repairs stenose; avoids dilation; allows consent; delay for the small vagina (**<3 cm**)
Hybrid	Clitorolabioplasty in infancy, vaginoplasty at puberty

KEY POINT

Both camps invoke oestrogen (maternal vs pubertal). Without comparative data the decision is ethical, not evidential; it rests with parents and team. Endocrine Society CAH guideline (2018): families with experienced surgeons decide; 2006 consensus limits surgery to Prader III–IV.

Body	Position
Lawson Wilkins and European paediatric endocrine societies' CAH group (LWPES/ESPE, 2002)	Surgery at 2–6 months; not between 12 months and adolescence
International intersex surgical society (ISHID) poll	Most prefer single-stage before 2 years
American Medical Association ethics council	Rejected a moratorium after 2 years' review
Advocacy groups; some US state bills	Moratorium until consent; bills not enacted

6 · Operative principles

Why this matters: moving the confluence wrongly causes incontinence.

Operate in this order: **clitoroplasty, then vaginoplasty, then labioplasty**.

Clitoral surgery	Technique	Verdict
Amputation; Lattimer recession	Excision; recession	Abandoned
Schmid (1961) MCQ	First corporeal excision preserving the neurovascular bundle	Basis of later techniques
Kogan (1983)	Subtunical excision, lateral Buck fascia incisions	—
Baskin (1999)	Defined neuroanatomy	**Ventral** incision spares the dorsal bundle
Pippi Salle (2007)	Corpora disassembled and buried in the labia	Reversible if male identity emerges

- The **Fortunoff (1964)** posterior perineal flap underlies almost every repair; it is now **omega-shaped**.
- **Hendren and Crawford (1969)** described the pull-through and showed that the **confluence level determines the type of vaginoplasty**.

Vaginoplasty	Indication	Principle	Pitfall
Cut-back	Labial fusion only	Incise sinus	Misses normal-calibre vagina
Flap	Low confluence	Posterior sinus opened, flap inset; confluence not moved	**Never for high confluence**: vaginal voiding, infection, incontinence
Pull-through	Very high	Vagina separated; sinus becomes urethra	If the vagina falls short, add flaps
Replacement	Absent vagina	*See Ch 58*	—

KEY POINT

Ask: **must the confluence be moved?** Flap brings skin to a low vagina; pull-through takes a high vagina off the urethra. Flap on a high confluence is the classic error.

7 · Low confluence repair

Why this matters: most CAH repairs are low.

ALGORITHM

1. Ensure metabolic stability; give **stress-dose steroids in CAH** and antibiotics.
2. Do endoscopy; place a **Fogarty catheter in the vagina** and a Foley in the bladder.
3. Prepare the whole lower body for supine–prone rotation (lithotomy blocks conversion).

ALGORITHM

1. Place a glans traction suture.
2. Incise around the glans, keeping the inner prepuce for the hood (**second only to the glans in sensitivity**).
3. Cut strips beside the ventral mucosal plate; raise the omega flap; make Y-incisions on the labia majora.
4. Deglove to the bifurcation, keeping the ventral plate intact.

- **Ventral subtunical excision**: incise Buck fascia on the **ventral-most** corpora and tease out the erectile tissue. Leave the tunics and dorsal bundle untouched, and ligate the stumps.
- If the bundle is mobilised, do it beneath the second layer of Buck fascia.
- Any **glans reduction** is ventral midline; never excise glanular epithelium. Fix the glans to the corporal stumps, not to the pubis.

ALGORITHM

1. Flap vaginoplasty: raise the omega flap with its fat.
2. Free the posterior sinus off the rectum (**hardest step**).
3. Open the posterior sinus in the midline into the posterior vaginal wall, and **continue to normal-calibre vagina**.
4. Inset the flap.

Labioplasty: split clitoral skin forms the **labia minora** (Marberger); the labia majora are moved down by **YV-plasty**.

KEY POINT

Two fixed points: never touch the dorsal neurovascular bundle; always open back to normal-calibre vagina.

8 · High confluence: pull-through

Why this matters: separating vagina from urethra risks the sphincter.

- Only ≈ **5%** of CAH cases are high; they are commoner in pure sinus.
- The **critical step** is separating the anterior vaginal wall from urethra and bladder neck. With no plane, sphincter and nerves are at risk, causing stricture, fistula or diverticulum.
- The **Passerini-Glazel flap** uses tubularised sinus and clitoral skin to build the anterior vaginal wall. **Gonzalez–Fernandes** uses a preputial flap.
- **ASTRA** (anterior sagittal transanorectal approach) is done prone, dividing the anterior rectal wall: excellent exposure, no colostomy, continence unaffected.

ALGORITHM

1. Position prone; incise in the midline between sinus and rectum.
2. Divide the sinus up to normal vagina and place a retractor inside the vagina.
3. Free the anterior wall off the urethra **under direct vision**.
4. Close the sinus as the urethra in 2–3 layers; mobilise the vagina.
5. Turn supine for flaps and labioplasty.

Err on the side of the vagina. If the vagina is short, add skin flaps (sinus tissue is more reliable). For a very small, very high vagina, consider waiting until puberty.

9 · Total and partial urogenital mobilisation

Why this matters: en-bloc mobilisation avoids that separation.

- **Total urogenital mobilisation (TUM, Peña 1997)** was devised for cloaca. The whole sinus is mobilised circumferentially (dividing the **avascular pubic ligaments**) to the perineum without separation.
- TUM gives less fistula and stenosis and better cosmesis. It **enables** vaginoplasty rather than replacing it.
- **Never discard the sinus**: split **ventrally** it becomes the vestibule (flap repair); split **dorsally** it becomes the anterior vaginal wall (pull-through).

	TUM	Partial (PUM, Rink)
Extent	Retropubic and suprapubic	Stops at the **pubourethral ligament**
Use	Higher confluence; cloaca	Most sinus repairs
Risk	Sphincter/nerve injury, stress incontinence	Lower; convert to TUM if needed

10 · Outcomes of sinus/DSD surgery

Why this matters: most girls need a second operation either way.

- After infant vaginoplasty, **secondary introitoplasty** is needed after puberty in **25–100%**, because the orifice does not grow; revision is easy.
- **Azziz (1986)** MCQ reported poor coitus rates, but almost all were **cut-backs**, so it does not apply now.
- Early single-stage repair gives less stenosis than late repair. **Confluence level, number of procedures and surgeon expertise** matter more than age; no data favour postpubertal surgery.
- Stenosis reaches over a third (cut-backs); dyspareunia is common. Orgasm is the same operated or not; sexual satisfaction matches controls except in **salt-wasting CAH**.
- **Elmasri meta-analysis (2018)**: mean age 2.7 years; female identity ≈ 89%; vaginal stenosis **27%**; only half had comfortable intercourse; no early-vs-late comparison.

Two-procedure dilemma	Infancy	After puberty
Approach 1	Simple clitorolabioplasty	Extensive vaginoplasty
Approach 2	Extensive single-stage clitorolabiovaginoplasty	Simple introitoplasty

Both usually mean two operations. The key factor is a **surgeon or team with a special interest**, which argues for centres of excellence.

11 · Cloaca — initial management

Why this matters: early decompression protects the kidneys.

- **Hendren's rule**: never do a separate rectal pull-through; repair rectum, vagina and urethra in one stage. Some groups stage it (anorectal first, urogenital peripubertally).

ALGORITHM

1. Decompress the bowel.
2. Decompress the genitourinary tract.
3. Correct nephron-destructive anomalies.
4. Then do the definitive repair.

Definitive repair is never the first operation.

Colostomy	Advantage	Drawback
Low loop	—	Bowel hard to obtain for vaginoplasty; interferes with pull-through
Right transverse, divided (Hendren)	Left colic supply intact; colon available for vaginal replacement	Absorbs urine, causing acidosis
Descending, divided (Levitt, Peña)	Less resorbing mucosa; enough distal colon; less prolapse	Leave a long sigmoid

- Urine absorbed by colon causes **hyperchloraemic acidosis**. **Preserve the appendix** for a later Malone antegrade continence enema (MACE).
- **Urinary decompression**: **CIC is first line** (even into the vagina), then a cloacal cut-back; use vaginostomy or vesicostomy only if CIC fails.
- **Kidneys**: structural anomalies occur in **60–83%**; chronic kidney disease (CKD) in **75%** and end-stage renal disease in **17%**. Urodynamics are abnormal in **90%**.
- Reflux is managed medically if urinary tract infection is prevented.

12 · Cloaca — definitive repair

Why this matters: channel and urethral length decide if TUM is safe.

Repair at **6–12 months**, after urinary evaluation and exclusion of hydrocolpos.

Common channel	Urethra	Procedure (Wood algorithm)
<3 cm	**>1.5 cm**	**TUM**
>3 cm	or <1.5 cm	Urogenital separation
Any	Any	Never transabdominal TUM (ischaemia)

KEY POINT

Why 1.5 cm? A urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags the bladder neck below the continence mechanism, which causes leakage.

ALGORITHM

1. **PSARVUP (Peña; posterior sagittal anorectovaginourethroplasty)**: place catheters in each channel; position **prone jackknife**.
2. Mark the anal site with a muscle stimulator; incise in the midline.
3. Open the rectal fistula and mobilise the rectum.
4. Do TUM or vaginal separation.
5. Close the urethra in 2–3 layers with a **fat flap**; rebuild the perineal body; centre the rectum in the sphincter.

- Rectum and vagina share a wall, so it is **better to enter the vagina than the rectum or urethra**. If the rectum lies above the pubococcygeal line, add laparoscopy or laparotomy.
- Vaginal separation is tedious, as the vagina often **almost encircles the urethra**; use **sharp dissection, not cautery**. The vaginal "switch" is abandoned.
- A very high or absent vagina needs bowel interposition *(see Ch 58)*.
- Aftercare: Foley for 2 weeks; rectal dilation from 2–3 weeks; colostomy closure at ≈ 3 months. Almost one-third empty poorly, so start CIC if in any doubt.

13 · Cloaca — results and follow-up

Why this matters: spine and channel length predict function.

- Outcome depends mostly on the **spine** (a short cord is worst) and **common-channel length**.
- **Urinary**: about half void spontaneously. Peña: CIC in **28%** (channel <3 cm) vs **78%** (>3 cm).
- Review yearly for life; a leak point pressure >40 cm H_2O is common despite a normal spinal MRI.
- **Renal**: CKD is mostly stage I–II and is preserved by aggressive bladder management.
- **Faecal** continence tracks neurology and improves into adulthood. A bowel programme comes first, then **MACE**.
- **Gynaecological**: introital stenosis, retained septum, fistula, deficient perineal body; about half of outside repairs need redo.
- Vaginal stenosis or obstructed menstruation occurs in **36–41%**. Pregnancy is possible, with **caesarean preferred**. Leave atretic genitalia alone.

> **ALGORITHM**
> 1. Do US **6–9 months after thelarche**.
> 2. Repeat every 6–9 months until menses are normal.

65 Adolescent and Transitional Urology

1 · Definitions

Why this matters: children with complex congenital conditions come to harm if nobody plans their move to adult care.

- **Adolescent urology** bridges care for complex congenital conditions. The adult urologist has operative skill but lacks **whole-patient knowledge** (prior operations, altered anatomy, demanding families).
- **Transition** (Got Transition) is the planned move from child/family-centred to adult/patient-centred care, in three stages: **planning, transfer and integration**. Transfer is one step.

KEY POINT

Evidence looks weak because the wrong endpoint is measured. Renal function and glycated haemoglobin (HbA1c) are expected to worsen, so **stability and retention in follow-up** are success; fewer are lost when actively transitioned.

2 · Who needs transition

Why this matters: the category decides how intense follow-up must be.

Category	Examples	Implication
1. Complex congenital	Exstrophy, posterior urethral valves (PUV), complex hypospadias, spina bifida (SB)	Lifelong care
2. Treated, now quiescent	Vesicoureteric reflux (VUR)	Early-adult review; may discharge with open return
3. Cured, with sequelae	Pelvic rhabdomyosarcoma	Late-effects surveillance
4. New adult concern	Fertility, pregnancy	Adult expertise

Failed transition (poor understanding, overbearing family, loss to follow-up, reliance on the emergency department, ED) raises morbidity and mortality because **preventive surveillance is lost**.

3 · The process

Why this matters: transition starts in early adolescence; it is not one referral letter at 18.

"Three Ps"	Content	Success measure
Preparation	Independence from 12–14 years; Transition Readiness Assessment Questionnaire (TRAQ)	All families involved
Plan	Clinic, imaging, blood-test schedule; self-care (catheterisation, irrigation, medications, whom to call)	Shared with patient and future team
Provider of the future	May or may not be a new team	Named provider

- The **paediatric provider must initiate**.
- The patient must be stable, so never transfer during a crisis. Exception: **pregnancy** needs expedited transfer with co-management.

- For **parents**, losing control is a major barrier. Give **private time** (sexual function), then ask what may be shared.
- Begin at about **12 years** (American Academy of Pediatrics, AAP); complete by the late teens–early 20s. Age **26** is the latest (US parental insurance ends). If independence is unachievable, plan custodianship.
- Programmes: Good 2 Go (Toronto), Ready Steady Go (UK), Got Transition (USA).

Got Transition core element	Age (years)
Policy	12–14
Tracking, readiness, planning	14–18
Transfer of care	18–21
Transition complete (integration into adult care confirmed)	18–23

> **KEY POINT**
> Transfer is not the endpoint: transition ends only when integration into adult care is confirmed.

4 · Routes and models of care

Why this matters: the route and model decide whether problems are caught early or only in an emergency.

- Routes: transitioned referral (ideal); referral without transition; a **new adolescent problem (ideal)**; and an **emergency after discharge**, which is the most challenging and often the most common.

Model	Strength	Drawback
1a Lifelong paediatric urologist	Continuity; suits limited life expectancy	Patient outlives the provider
1b Paediatric urologist in adult department	Continuity and adult expertise	Cost, hospital privileges, geography
2 Transfer to an adolescent/adult specialist	Specialist coordinates the multidisciplinary team (MDT)	Cross-institution communication
3 Direct transfer, no plan	The default	Invisible until an emergency

Lifelong and multidisciplinary need are **independent**: exstrophy may need only a urologist, whereas SB needs the most complex MDT. **Nursing** is the backbone.

5 · Development and consent

Why this matters: development guides surgical timing; capacity guides consent.

- Avoid genital surgery in the phallic–oedipal period (**3–6 years**) and separation in the first year. Theory and the 1975 AAP panel favour genital surgery at **<6 months**.
- Augmentation is withheld until the child can catheterise and wants dryness.
- **Emerging adults (18–25 years)** consent for themselves unless they lack capacity. Self-management matures into the third decade.
- Patients with **SB** usually have normal intelligence but deficits in **working memory and executive function**.
- **Medical post-traumatic stress disorder (PTSD)** causes distrust, avoidance and procedural panic; cognitive-behavioural therapy (CBT) may help.

6 · Setting up a service

Why this matters: a transition clinic needs the right team and institutional backing.

- Core MDT: paediatric and adult urologists, specialist nurses, care coordinator, nephrologist, psychologist. Gain support with a **business model**.
- Consent and confidentiality are reviewed **with the family present, then with the patient alone**. Standing orders (for example, urine cultures) avoid ED visits.

Barrier	Solution
Patient/parent: attachment, fear	Address the child directly early; readiness questionnaires
Paediatric provider: family trust	Ask "Am I the best person?"
Adult provider: **unfamiliarity with congenital conditions (biggest)**; finance	Partnerships, networks, training

120 Aging and Geriatric Urology

1 · Concepts and biology

Why this matters: function matters more than age.

- **Chronological age is a poor descriptor**; physiological and functional age (function, frailty, geriatric syndromes) matter.
- "Geriatrics" was first used clinically by Nascher (1909) MCQ.
- **Cellular senescence** (irreversible arrest of replication) drives ageing. It is replicative, via telomere shortening (**Hayflick limit**, 1961) MCQ, or stress-induced (reactive oxygen species, radiation, DNA damage).
- The **senescence-associated secretory phenotype (SASP)** (cytokines, growth factors, matrix metalloproteinases) causes chronic inflammation and tumorigenesis and ages neighbouring tissue.
- **Healthy ageing** (World Health Organization): maintaining functional ability; target function, not absence of disease.

2 · Clinical evaluation

Why this matters: function and frailty predict surgical outcome better than age.

- Geriatric syndromes act **synergistically** (many factors, one phenotype). **Homeostenosis** is narrowed physiological reserve, so recovery after a stressor is poorer.

Tier	Content	Significance
Basic activities of daily living (ADLs)	Bathing, dressing, grooming, toileting, feeding, ambulation	**Bathing lost first**, feeding last; loss predicts dependency and postoperative outcome
Instrumental ADLs (IADLs)	Cooking, cleaning, transport, finances, telephone, medicines	Independent living

- Function is lost top down; each step shortens survival. Screen: **can the patient bathe independently?**
- **Surgical risk**: age is seldom independent; predictors are **longer operating time, poor American Society of Anesthesiologists (ASA) score and urgent surgery**. Optimise, do not "clear"; this helps even >80.

ALGORITHM

1. Screen for frailty before elective major surgery (Society for Perioperative Assessment and Quality Improvement, SPAQI).
2. If the screen is positive, do a diagnostic frailty assessment.
3. Then do a comprehensive geriatric assessment (CGA).

- The **American College of Surgeons National Surgical Quality Improvement Program (ACS NSQIP)/American Geriatrics Society (AGS) checklist**: 13 items (cognition/capacity, depression, delirium risk, substance use, cardiac and pulmonary risk, function/falls, frailty, nutrition, medications, goals, social support, tests); only 2 are organ systems.
- **CGA** (geriatrician-led): frailty, physical performance, albumin, creatinine, brain natriuretic peptide (BNP) and forced expiratory volume in 1 second (FEV1).

3 · Geriatric syndromes

Why this matters: frailty and polypharmacy cause much harm.

- Frailty, depression, delirium, falls, incontinence, polypharmacy, sensory loss and pressure ulcers feed one another (frailty, urinary incontinence [UI], falls), so treating one helps others.

Frailty is decreased reserve causing vulnerability; it is distinct from comorbidity and disability.

	Accumulated deficits index	Fried phenotype
Method	Count failing systems	(1) weight loss >4.5 kg or 5%/year; (2) weak grip; (3) slow gait; (4) low activity; (5) exhaustion
Cut-off	Continuous	**≥3 frail; 1–2 pre-frail**
Validity	Does not predict postoperative outcome	Predicts death, dependence, falls, postoperative complications

Polypharmacy — Beers criteria

Harm often comes via the **prescribing cascade** (a side effect treated with another drug).

Drug	Concern and action
Antimuscarinics (in dementia/delirium), tricyclics, first-generation antihistamines	Anticholinergic burden: delirium, falls, retention. Avoid combinations; no tricyclics in benign prostatic hyperplasia (BPH)
Non-steroidal anti-inflammatory drugs (NSAIDs)	Gastrointestinal bleeding; renal and heart failure at creatinine clearance (CrCl) <30 mL/min
Systemic oestrogen; testosterone	Cancer, cardiovascular harm. **Vaginal oestrogen safe**; testosterone only for confirmed symptomatic hypogonadism
Ciprofloxacin, macrolides (**not azithromycin**), co-trimoxazole	Warfarin potentiation: monitor international normalised ratio (INR); dose-reduce at CrCl <30
Benzodiazepines	Cognition, gait, delirium. Avoid with opioids or ≥2 other central nervous system drugs
Nitrofurantoin	Listed at CrCl <30/chronic use; usable in moderate impairment
Desmopressin; tramadol	Hyponatraemia: treat nocturia's cause; check sodium

> **KEY POINT**
> Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance.

Genitourinary syndrome of menopause (GSM)

- Coined 2014; a cause of transient incontinence. Dryness, dyspareunia, urinary tract infection (UTI) **plus urgency and urgency urinary incontinence (UUI)**. Sex-steroid loss drives overactive bladder (OAB) in both sexes, including men on androgen deprivation therapy (ADT) or 5α-reductase inhibitors (5-ARIs).
- Treat with **vaginal oestrogen** (or moisturisers, dehydroepiandrosterone [DHEA], ospemifene). Prior breast/endometrial cancer is not an absolute bar; consult the oncologist.
- **Urethral caruncle**: painless spotting, so exclude endometrial/cervical cancer. Topical oestrogen; excision rarely.
- Faecal incontinence is under-reported: **ask specifically**.

4 · Incontinence management

Why this matters: incontinence drives nursing-home admission; its drugs can harm cognition.

- Common but not normal; behavioural therapy first *See Voiding Dysfunction book, Ch 110*. †Transient causes (DIAPPERS): delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess output, restricted mobility, stool impaction.
- P2 P3 Overactive bladder and its drug therapy; OAB Pharmacotherapy — *trospium quaternary, M3 selectivity, tolterodine QT, mirabegron BP/metoprolol, Beers avoid in dementia*

Not in the paper answer:

- The paper suggests fesoterodine for the elderly, but Beers lists it (below) as "avoid" in dementia.
- **Antimuscarinics**: fewer than half persist at 6 months. The ageing blood–brain barrier is leakier, which causes cognitive harm.
- High-dose tolterodine QT prolongation risks torsades de pointes.
- Retention is uncommon, but check the post-void residual (PVR) after starting and never exceed the maximum dose.
- Beers "avoid" in dementia: **oxybutynin, fesoterodine**; trospium has the least central penetration.

Nursing homes

- UI affects three-quarters of long-stay residents and drives admission. Catheter colonisation is **98%**, so treating asymptomatic bacteriuria (ASB) causes harm.

Loeb — catheter	Loeb — no catheter	National Healthcare Safety Network (NHSN)
≥1 of: fever, new costovertebral tenderness, rigors, **acute mental status change**	Acute dysuria; or fever plus new urgency, frequency, suprapubic pain, haematuria or UI	Culture ≥10^5 colony-forming units (CFU), no other source, and clinical signs

> **KEY POINT**
> A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used.

5 · UTI and asymptomatic bacteriuria

Why this matters: treating bacteriuria without symptoms harms without benefit.

- Separate UTI from OAB and ASB. **A causal link between confusion and UTI is not supported** *See Infections book, Ch 26*.
- ASB: up to 22% of older women, ~10% of older men; no antibiotics. The US Preventive Services Task Force (USPSTF) gives grade D against screening. Infectious Diseases Society of America (IDSA): treat **only in pregnancy and before urological instrumentation**.

6 · Cancer, end of life

Why this matters: fitness, not age, decides cancer treatment.

- **Prostate**: USPSTF grade D, so no prostate-specific antigen (PSA) screening >70. With life expectancy **<10 years**, plan non-curative care.
- **Bladder**: restaging transurethral resection (TURBT) for T1 is often omitted in the elderly yet improves survival. Bacille Calmette–Guérin (BCG) is less effective (immunosenescence). Cystectomy at ≥75 is comparable with careful care.

- Urologists must report **elder mistreatment** in most places.
- **Palliative care is not withdrawal**: aggressive symptom control; goals-of-care discussion first.
- Age-Friendly "4 Ms": Mobility, Medication, Mentation, what Matters.

> **KEY POINT**
> Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide.

7 · Landmark trials

Why this matters: which prevention helps older adults.

Trial	Design	Result	Take-home
ASPREE	Randomised controlled trial (RCT): aspirin vs placebo, healthy elderly	No disability-free survival gain; higher mortality	No primary-prevention aspirin
Testosterone Trials	RCTs, hypogonadal older men	No cognitive benefit; modest sexual-function gain	Not for cognition

Effective prevention: calcium and vitamin D (fractures), antihypertensives (mortality even >80), statins, exercise.

126 Management and Considerations in Transgender and Nonbinary Individuals

1 · Definitions and terminology

Why this matters: correct, respectful terms build trust and prevent clinical errors.

- **Transgender and gender-diverse (TGD)**: gender identity differs from sex assigned at birth. Four **independent axes** exist (assigned sex, gender identity, gender expression, sexual orientation); most errors collapse one into another.

Term	Meaning
Gender identity / expression	Internal sense of gender / its outward display
AFAB / AMAB	Assigned female/male at birth (by genitalia and/or genetics)
Cisgender	Identity matches assigned sex
Transgender man / woman	Male identity, AFAB / female identity, AMAB
Transmasculine / transfeminine	Direction of gender, not a binary endpoint
Nonbinary; agender; gender fluid	Neither male nor female; no gender; changes over time
Gender dysphoria	Distress from identity–assigned sex discrepancy
Gender incongruence	Diagnostic term: marked, persistent incompatibility
Sexual orientation	Attraction; cannot be inferred from identity
"Transsexual"	Historical; disfavoured

- Not all TGD people have dysphoria; some need only social transition, others gender-affirming hormone therapy (GAHT) and/or gender-affirming surgery.

2 · Diagnosis and assessment

Why this matters: diagnosis and consent standards decide who starts treatment.

	DSM-5 (2013)	ICD-11 (2019)
Term	Gender dysphoria	**Gender incongruence** (replaced "gender identity disorder")
Criteria	Marked incongruence for ≥ **6 months**, significant distress or impairment, not solely due to prejudice	Marked, persistent incongruence; desire for characteristics/treatment of experienced gender
Chapter	Mental disorders	**Moved to Conditions Related to Sexual Health** MCQ

- (DSM-5: Diagnostic and Statistical Manual of Mental Disorders, 5th edition; ICD-11: International Classification of Diseases, 11th revision.) ICD-11 keeps a code for access but drops the mental-disorder label. **Severe distress is not a prerequisite**; marked, sustained incongruence is.

WPATH Standards of Care 8 (SOC-8, 2022)

- World Professional Association for Transgender Health (WPATH): treat only marked, sustained incongruence; exclude other causes; optimise coexisting conditions.

- Assess **capacity to consent**, and give **fertility counselling before any treatment**, not just surgery.
- **A single opinion** from a competent professional suffices for medical and surgical treatment.
- Before gonadectomy, **consider ≥6 months of hormones** unless not indicated (a consideration, not a rule).

> **KEY POINT**
> SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling stay firm because consequences are irreversible.

3 · Gender-affirming hormone therapy (GAHT)

Why this matters: urologists see GAHT complications and adjust it after gonadectomy.

- GAHT is safe under supervision; the aim is **physiological** levels of the affirmed gender. Eligibility: persistent documented dysphoria, capacity, age of majority and controlled comorbidities.

Direction	Agent	Dose
Transfeminine — oestrogen	Oral oestradiol	2–6 mg/day
	Transdermal oestradiol	0.025–0.2 mg/day
	Oestradiol valerate/cypionate intramuscular (IM)	5–30 mg 2-weekly or 2–10 mg weekly
Transfeminine — antiandrogen	Spironolactone	100–300 mg/day
	Cyproterone acetate	25–50 mg/day
	Gonadotropin-releasing hormone (GnRH) agonist	3.75 mg monthly / 11.25 mg 3-monthly
Transmasculine — testosterone	Enanthate/cypionate	100–200 mg 2-weekly
	Undecanoate	1000 mg 12-weekly (rare in the US: oil pulmonary embolism)
	Gel 1.6% / patch	50–100 mg/day / 2.5–7.5 mg/day

> **KEY POINT**
> **Why the asymmetry?** Physiological oestrogen alone does not suppress testosterone to the female range, so transfeminine therapy needs two components. Testosterone suppresses the ovarian axis on its own, so transmasculine therapy needs one.

Transfeminine

- **Ethinyl oestradiol is not recommended** (~20-fold thrombosis; cannot be measured). Use oral 17β-oestradiol, target **100–200 pg/mL**. **Use transdermal if there is venous thromboembolism (VTE) risk**: age >45 or previous VTE.
- **Spironolactone** (US commonest) blocks the androgen receptor and partly inhibits 17α-hydroxylase/17,20-lyase. **Cyproterone** is the European standard. Alternatives: GnRH agonists, bicalutamide. 5α-reductase inhibitors are weaker; progesterone is unproven.
- Effects take 3–12 months. **Irreversible: breast development, reduced or absent fertility.** Risks: VTE, cardiovascular disease (CVD), hypertriglyceridaemia, liver dysfunction, osteoporosis and hormone-related cancer.

- Monitor levels 3-monthly in year 1, then 1–2 times a year. Target testosterone **<50 ng/dL**. Check electrolytes/creatinine on spironolactone, and prolactin; stop smoking.
- **After gonadectomy, reduce oestrogen by 50–75%** (titrate to luteinising hormone [LH], follicle-stimulating hormone [FSH] and bone density) and **stop the antiandrogen**.

Transmasculine

- Contraindications: pregnancy, unstable coronary disease, untreated polycythaemia. Oestrogen-sensitive cancer needs oncology input, as testosterone aromatises. **Contraception is essential** because testosterone is teratogenic.
- Target testosterone **300–1000 ng/dL**. Persistent menses: progestin or endometrial ablation.
- At 1–2 months: menses stop, acne, clitoromegaly, vaginal atrophy, higher libido; hair, muscle and voice change later. **Irreversible: clitoral growth, facial hair, voice change, male-pattern baldness.** Menses return 2–6 months after stopping.
- Risks: **erythrocytosis**, liver dysfunction, sleep apnoea, CVD. Monitor testosterone and **haematocrit** 3-monthly in year 1, then 1–2 times a year.
- **After gonadectomy, testosterone dose is not reduced**.

Nonbinary regimens are individualised. AFAB: a progestin, intrauterine device or hysterectomy, with testosterone and/or **nandrolone** (muscle without much hair). AMAB wanting to keep erections: oestrogen with or without a low-dose antiandrogen.

4 · Urological considerations

Why this matters: urinary symptoms are common, and surgery is not always the cause.

Lower urinary tract symptoms (LUTS) have five contributors: **natal anatomy, hormones, surgery, pelvic floor dysfunction and psychosocial/behavioural factors**.

- **Testosterone-induced vaginal atrophy** (dryness, bleeding, dyspareunia, urgency, frequency) is treated with topical vaginal oestrogen.
- **After vaginoplasty**, about a third report altered voiding (**nocturia is commonest**). Spraying, transient incontinence, meatal stenosis, stricture and urethroneovaginal fistula also occur *See Ch 127*.
- **After phalloplasty or metoidioplasty**, post-void dribble comes from urine pooling at the junction of the native urethra and pars fixa; compressing the scrotal base empties it. Stricture and fistula are far commoner *See Ch 128*.
- Treat **pelvic floor** dysfunction before surgery; this eases neovaginal dilation.
- **Tucking** (hiding the genitals) causes stasis, prostatitis-like symptoms, epididymo-orchitis and urinary tract infection (UTI).
- Exclude atrophy, tucking and pelvic floor causes before blaming the operation.

5 · Fertility preservation

Why this matters: GAHT and gonadectomy can end fertility permanently.

- Counsel **before hormones and/or surgery** (WPATH, American College of Obstetricians and Gynecologists, American Society for Reproductive Medicine, Endocrine Society). Barriers: cost, urgency, provider bias, process-provoked dysphoria.

	Transfeminine	Transmasculine
Standard	**Sperm cryopreservation**; hold GAHT for ≥ **3 months** before collection	**Oocyte or embryo cryopreservation** after ovarian stimulation
Prepubertal, or cannot pause hormones	Testicular tissue cryopreservation (experimental)	Ovarian tissue; needs no hormone pause or stimulation
Obstacles	Ejaculation distress: vibratory stimulation, electroejaculation or surgical retrieval	Bleeding returns; pelvic examination and transvaginal ultrasound cause distress

Off testosterone, menses return in 3–6 months, and most conceive within 6 months.

6 · Cancer screening — screen the organ, not the gender

Why this matters: retained natal organs keep their cancer risk.

- **Prostate (transfeminine)**: rare but may be aggressive (GAHT begun after 50). Shared decision-making on prostate-specific antigen (PSA) at 55–69. **On GAHT, the upper limit is 1 ng/mL, or a rise of ≥ 0.3 ng/mL from nadir**, †because androgen suppression lowers PSA. After full-depth vaginoplasty, feel the prostate through the **anterior neovaginal wall**. Biopsy is transrectal, perineal or transneovaginal.
- **Testis**: only case reports, so self-examination only.
- **Cervix (transmasculine)**: cytology 3-yearly at 21–29 years; at 30–65, cytology 3-yearly or high-risk human papillomavirus (hrHPV) testing/co-testing 5-yearly. Stop if there is no cervix, or at >65 after two negatives. Use a small speculum and **vaginal oestradiol for 1–2 weeks before**.

7 · Affirming practice

Why this matters: patients avoid care after poor experiences.

- Provide lifelong organ-appropriate screening, a **detailed organ inventory** and inclusive intake forms (name, pronouns).
- **Trauma-informed care** (Substance Abuse and Mental Health Services Administration, SAMHSA) uses the "Four Rs": **Realise, Recognise, Respond, Resist retraumatisation**.

127 Male-to-Female Gender-Affirming Surgery: Vaginoplasty

1 · Aim and preparation

Why this matters: many failures start with poor preparation, above all a patient who cannot dilate.

- Surgery aligns anatomy with identity **and** preserves voiding, sensation and sexual function (terminology, hormones: *see Ch 126*).
- **Mental health** input treats dysphoria and judges **surgical readiness**; start early (housing, support, smoking, weight).
- Key risks: **failure to dilate**, worsening psychiatric prognosis. Dysphoria-driven depression or suicidality should **improve**, so it is no reason to defer.

Older criteria	World Professional Association for Transgender Health (WPATH) Standards of Care 8 (SOC8)
Two assessments	**One** (inter-assessor agreement is high)
Mental health professional only	Any licensed clinician ≥ master's degree, competent in transgender health (incl. primary care or surgeon)
Hormones before gonadectomy	**6 months** suffices (experience testosterone blockade before irreversible change)
Comorbidity "well managed"	Risks and benefits discussed
12 months' "real-life experience"	**Removed**: dangerous for those facing violence

- **Capacity is procedure-specific**; if doubtful, a psychologist plans support, with or without proxy decision-makers.

> **KEY POINT**
> SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory.

- Stay within **1 hour's drive** of hospital for the first **3 weeks**; record affirmed name and gender; confirm support for dilation.
- **Body mass index (BMI)** mainly determines wound healing; no cut-off is proven, many require **≤35 kg/m²**; a higher BMI may suit vulvoplasty.
- **Permanent preoperative hair removal** of the scrotal–perineal graft template: intraoperative follicle removal fails (phased growth), and canal hair causes infection and hairballs.
- **Continue oestrogen perioperatively**: a 2022 meta-analysis found no added venous thromboembolism (VTE) risk.

Orchiectomy

- It may be the sole operation or the first stage; afterwards essentially all stop spironolactone. Offer **fertility counselling and sperm banking to all**.
- Use a **vertical midline scrotal incision**: an inguinal one may compromise penile flap supply, while a scrotal one is incorporated into the later graft.

KEY POINT

Plan every orchiectomy as though vaginoplasty will follow: use a midline scrotal incision and leave peri-cord fat for labial bulk.

2 · Penile inversion vaginoplasty (PIV)

Why this matters: PIV is the commonest technique; serious injuries occur in the prostate–rectum canal dissection.

- P2 P4 Penile inversion vaginoplasty technique — *homologous tissue table, operative steps, dilation, complication rates*

Not in the paper answer:

- Late bulbospongiosus excision, vulvoplasty after prostatectomy, stricture rate 1–6% (all below).
- Physiotherapy starts preoperatively; a speculum checks depth and granulation at follow-up.
- Prolapse reflects poor graft take, not pelvic floor weakness; stenosis often follows repacking.
- Set-up: lithotomy, calf and heel padded (**common peroneal nerve**), heparin and compression. Give **tranexamic acid 1 g IV at start**, as bleeding is front-loaded.

ALGORITHM

1. Foley and drain; mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion).
2. Harvest scrotal graft and small posterior rhomboid flap (relieves introital tension); defat, tubularise.
3. Orchiectomy, leaving cord fat; penile disassembly; corporal excision and clitoroplasty.
4. Urethroplasty, then canal dissection; line canal, labiaplasty, pack.

- **Disassembly**: dartos stays with skin. The neurovascular bundle is best **left on the dorsal tunica** (less nerve damage). Open the corpora ventrally, coagulate the cavernosal arteries and **excise all erectile tissue**.
- **Clitoroplasty**: a "W" on the proximal glans keeps the lateral wings. Excess glans looks penis-like. A short penile shaft shortens the labia minora, not the canal.
- **Urethroplasty**: excise bulbospongiosus **only after canal dissection**, as it covers a rectal injury. Spatulate downward for a seated stream.
- **Canal**: divide the central tendon; use a Lowsley retractor and repeated digital rectal examination. **Preserve Denonvilliers fascia** and close it if breached. A prostatic capsule breach means checking the urethra. **Packing** ensures graft take.
- Keep the labia majora incision **≥1 cm** from the groin crease (dehiscence). Finish with an antibiotic pack and negative-pressure dressing.

Vulvoplasty (zero-depth)

- It suits those not wanting receptive intercourse and avoids **lifelong dilation** and rectal or bladder injury. It is not freely reversible (graft discarded, scarring).
- **Prior radical prostatectomy favours vulvoplasty** (sphincter injury risk).

3 · Abdominoperineal and intestinal options

Why this matters: when skin is short or the plane risky, peritoneum or bowel can line the canal.

- **Peritoneal (Davydov-type)**: the proximal canal is made robotically, and rectal injury fell from **~7%** historically to ~2.3% (urethral 1.1%). The **rectum may mimic seminal vesicles**;

enter between the Denonvilliers leaflets.

- Flaps: **two-flap** (bladder and sigmoid peritoneum, 10 × 10 cm each) or **pull-through** (abdominal wall, 11 × 20 cm, on deep inferior epigastrics). **Close peritoneal defects** (hernia).
- **Intestinal** vaginoplasty suits severe penoscrotal hypoplasia or revision stenosis. Contraindications: inflammatory bowel disease (IBD), adhesions, irradiation, short gut. Problems: mucus, diversion colitis, prolapse; ~1 in 8 removed. **Dilation is still required.**

KEY POINT
Lining ladder: penile and scrotal skin first, then peritoneum, then bowel. Each adds depth and lubrication but imports donor pathology; none removes the duty to dilate.

4 · Complications

Why this matters: rectovaginal fistula is the most feared; stenosis is the lasting threat.

Complication	Cause / key fact	Management
Wound dehiscence (commonest minor)	Introitus, dilator tension	Local care; granulation: curette or cautery plus steroid
Rectovaginal fistula (most devastating)	**0.8%** primary, 6.3% revision	Intra-op: watertight closure and bulbospongiosus flap, with or without diversion. Fistula: stop dilating, let canal stenose, revise later, **or** delay closure
Urethrovaginal fistula (0.8–3.9%)	Missed injury, distal stenosis	Distal (vaginal voiding): often left. Bladder-neck (incontinence): transvaginal repair; Martius flap unusable
Urethral stricture (1–6%)	Split stream, spraying, urinary tract infection (UTI), retention	Exclude pelvic floor dysfunction; dilate if short, else meatal spatulation
Urethral bulge / upward meatus	Under-debulked spongiosum	Debulk, fix meatus downward
Vaginal stenosis	Poor dilation, scar, graft loss, pelvic floor dysfunction	Pelvic floor physiotherapy **first**; then scar excision and graft, or peritoneal/bowel augmentation
Prolapse (4–10%)	Poor graft take or fixation	Early repack; peritoneal fixation acts as a colpopexy

KEY POINT
Three failure modes: violated plane (fistulas), tissue not excised (bulge, engorgement), graft not taken or maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient — dilation is part of the operation.

5 · Outcomes and follow-up

Why this matters: lifelong dilation keeps the result, and the retained prostate still needs surveillance.

- **Median time to orgasm is ≈ 180 days**. Depth is 12–16 cm (cisgender 9–12 cm); self-lubrication is not the norm.
- Stay is ~**5 days**; on **day 5 pack and catheter come out** and dilation is taught. Retention: recatheterise for 1 week.

- Follow-up: weekly × 3, then 6 weeks, 3, 6, 12 months, annually. †The prostate is retained; survey it via the anterior neovaginal wall.

Period	Dilation (15–20 min each)
0–6 weeks	4 × daily
6 weeks–3 months	3 × daily
3–12 months	2 × daily
>1 year	Daily; less if stable

KEY POINT

The body treats the canal as a defect to close, so stenosis is a lifelong risk. Any narrowing means more dilation, contacting the surgeon and physiotherapy.

128 Special Urologic Considerations in Transmasculine Individuals

- P4 Enlist the steps in female to male gender reassignment surgery — *staged sequence, vaginectomy, neourethra parts, metoidioplasty vs phalloplasty, scrotoplasty, prostheses, complication sites*

Not in the paper answer:

- Haemorrhage is the commonest vaginectomy complication; two-stage repair is standard for long strictures; never repeat urethrotomy.
- For retention: Coudé, then stylet, then cystoscopy; emergency suprapubic catheter.

1 · Scope

Why this matters: genital surgery always alters the urinary tract.

- Diagnosis, World Professional Association for Transgender Health (WPATH) standards, hormones: *see Ch 126*. Goals: male appearance, **voiding standing**, sensation and penetration.
- **Early puberty blockade has a surgical price**: an oestrogen-naive vagina is harder to remove, and small labia majora give less scrotal tissue. Avoiding public toilets may cause an underactive detrusor.

2 · Preoperative care and consent

Why this matters: urinary problems are far harder to fix after urethral lengthening.

- **Urological screening is mandatory**: uroflow, post-void residual (PVR), renal ultrasound. Treat lower urinary tract dysfunction (LUTD), stones, hydronephrosis first.
- WPATH Standards of Care 7 (SOC7) requires ≥**12 months** of testosterone before genital surgery (†SOC8: 6 months) for **clitoral hypertrophy**, which plateaus at ≈2 years.

Agent	Helps	Hinders
Testosterone	Clitoral hypertrophy aids metoidioplasty, lengthening	Vaginal atrophy: harder vaginectomy
Gonadotrophin-releasing hormone (GnRH) agonist	Prevents female characteristics	Small labia majora: less scrotoplasty tissue

- Consent: fistula, stricture; vaginectomy bleeding, mucocele, rectal/bladder/sphincter injury; LUTD, urinary tract infection (UTI); implant problems.

3 · Removal of female structures

Why this matters: vaginectomy is hazardous but required for urethral lengthening.

- The ureter lies beneath the uterine artery; injury needs immediate repair. †If the cervix is retained, screening continues.
- **Vaginectomy** risks: haemorrhage, then bladder, then rectal injury. **Without vaginectomy, no urethral lengthening** (fistula). Alternative: fulgurate upper vaginal mucosa (prevents mucocele).

ALGORITHM

1. Hydrodissect (dilute lidocaine); circumferential introital incision.
2. Blunt rectovaginal plane; urethrovaginal plane **sharply, with no diathermy** (protects sphincter), then the bladder plane.
3. If bladder injured: close, with or without cystoscopy.
4. Free levators, control paravaginal plexus; approximate levators (prevents prolapse).
5. Bulbospongiosus over urethra; layered closure.

Timing	Verdict
All-in-one	More major complications, transfusion with phalloplasty; none extra with metoidioplasty
Early vaginectomy, phalloplasty later	Keeps blood loss away from free-flap perfusion (theoretical)
Two-stage (preferred)	Mastectomy, hysterectomy–oophorectomy; then vaginectomy, genitoperineal reconstruction

4 · Genital reconstruction

Why this matters: the neourethra lacks spongiosum, which explains most complications.

- The fixed (perineal) urethra is like a bulbar urethra, and the phallic part like a penile one, both without spongiosum. One **dorsal clitoral nerve** (1 or 11 o'clock) is coapted for sensation.
- **Bulbospongiosus over the fixed part** gives fewer fistulas and expels last drops. Anterior vaginal wall flaps are avoided (necrosis, sacculation).
- Scrotoplasty flaps run on the **external pudendal artery**. Phalloplasty has many flaps, none ideal.

	Metoidioplasty	Phalloplasty
Phallus	Own clitoris, **3–8 cm**	Flap
Erection	Spontaneous	Needs prosthesis
Sensation	Largely kept	Depends on nerve coaptation
Penetration	Usually not	Principal goal
Donor scar	None	Visible

5 · Postoperative care and catheterisation

Why this matters: forceful catheterisation can ruin the neourethra.

- Suprapubic (SPC) and neourethral catheters are placed; the neourethral one comes out at **14–21 days**. Clamp the SPC, recording PVR until voiding is normal.
- Catheterisation is hard: the curve into the native urethra is abrupt; Bartholin-duct sinuses catch the tip.

ALGORITHM

1. One gentle Coudé attempt, then one stylet-guided attempt.
2. If any difficulty, stop: flexible cystoscopy, guidewire, open-tip catheter.
3. In an emergency, place a **suprapubic catheter**.

6 · Urethral complications

Why this matters: fistula and stricture drive most revisions.

- **Fistula** is commonest at the phallic base. Keep diversion; small ones close. If persisting **>3 months**, excise, close in layers with fat interposition, catheter 10–12 days.
- **Stricture** is commonest at the **fixed-phallic anastomosis**, also the meatus. A plateau uroflow prompts retrograde urethrography (RUG). Treat after **≥6 months**; retention needs SPC or perineal urethrostomy.

Option	Use
Dilation	Low-grade meatal stenosis; not curative for anastomotic
Direct vision internal urethrotomy (DVIU)	Cold knife over guidewire, catheter 2 weeks; low success, never repeat
Meatotomy / Heineke–Mikulicz	Short meatal / short non-obliterative
Excision and primary anastomosis	Short, near-obliterative anastomotic
Free graft	Needs flap support (fat bed) or dorsal inlay on clitoral corpus
Two-stage (standard)	Long/complex; retubularise ≥6 months later
Flap / perineal urethrostomy	Recurrent; or no further reconstruction wanted

- **Hair**: prevent with a hairless flap or laser. Bezoar or stones: open, remove, epilate, retubularise later.
- **Post-void dribbling** is near-universal (no spongiosum); use perineal pressure and milking. If severe, exclude sacculation or a vaginal-remnant diverticulum.

> **KEY POINT**
> The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure.

7 · Prostheses

Why this matters: implants fail often, so timing matters.

- Erectile prosthesis: ≥**1 year** after phalloplasty, with **protective sensation** and a stable urethra. One cylinder is anchored to pubic periosteum (no tunica); malleable is simpler.
- Frequent infection or erosion needs urgent explantation; reimplant ≥6 months later.

★ Putting it together

A · Newborn with atypical genitalia

Why this matters: the first threat to life is adrenal crisis.

ALGORITHM

1. **Emergency is adrenal:** congenital adrenal hyperplasia (CAH) is commonest (half of all; 90% of 46,XX). Salt wasting (low Na, high K, shock) appears at day 5–7 (*see Ch 59*).
2. **Tests:** karyotype (SRY), 17-hydroxyprogesterone after 36 h, electrolytes, pelvic ultrasound for uterus.
3. **Team:** †defer sex assignment until multidisciplinary team evaluation.
4. **Gonads:** if dysgenetic with Y (e.g. complete gonadal dysgenesis, 30–45% tumour risk), then prophylactic gonadectomy.

B · Urogenital sinus and cloaca

Why this matters: anatomy decides the operation.

KEY POINT

True urethral length (bladder neck to confluence) outweighs common-channel length. Low confluence needs a flap; high needs pull-through or total or partial urogenital mobilisation. Clitoroplasty only for Prader III–IV, sparing the dorsal bundle. Cloaca: colostomy, decompress hydrocolpos, cloacogram or MRI with endoscopy, then posterior sagittal repair; seek VACTERL and tethered cord (*see Ch 60*).

C · Absent vagina

Why this matters: karyotype and uterus sort the causes; dilation comes first.

Cause	Key feature	Management
Mayer–Rokitansky–Küster–Hauser	46,XX, normal ovaries and breasts, no uterus; †image kidneys	Dilation first; else McIndoe, bowel, buccal, peritoneal
Complete androgen insensitivity	46,XY, testes, no uterus	Dilation usually suffices (*see Ch 59*)
Transverse septum / imperforate hymen	Uterus present, haematocolpos at puberty	Hymen: transverse incision, not aspiration; septum: excision (*see Ch 58*)

D · Immunotherapy and hereditary cancer

Why this matters: biology now picks the drug.

ALGORITHM

1. **Checkpoints:** tumour interferon-γ induces programmed death ligand-1 (PD-L1), silencing PD-1-positive T cells; CTLA-4 inhibitors act earlier, at priming. †Immune-related adverse events (colitis, hepatitis, pneumonitis, endocrinopathy): if grade ≥2, hold drug, give steroids; endocrinopathies need lifelong replacement (*see Ch 24*).
2. **Hereditary:** von Hippel–Lindau (VHL; clear cell; belzutifan), hereditary leiomyomatosis and renal cell cancer (fumarate hydratase; aggressive papillary), Birt–Hogg–Dubé (FLCN; hybrid oncocytic), Lynch (mismatch repair; upper-tract urothelial cancer).
3. BRCA or homologous-repair-deficient prostate cancer: PARP inhibitor. FGFR-altered urothelial cancer: erdafitinib (*see Ch 25*).

E · Older patient before surgery

Why this matters: frailty predicts complications.

KEY POINT

Function, not age, predicts outcome. Screen frailty, then comprehensive geriatric assessment: cognition, delirium, nutrition, falls, polypharmacy. Avoid anticholinergic burden (β3-agonist for overactive bladder). New incontinence: transient causes first (†DIAPPERS) (*see Ch 120*).

F · Transgender and nonbinary patients

Why this matters: screen the organs present.

ALGORITHM

1. Organ inventory guides screening: retained prostate after vaginoplasty (via anterior neovaginal wall; †PSA is lower on oestrogen), retained cervix on testosterone (*see Ch 126*).
2. Vaginoplasty: stenosis, rectovaginal fistula (most devastating), urethrovaginal fistula (*see Ch 127*).
3. Phalloplasty: fistula and stricture; without vaginectomy, more fistula (*see Ch 128*).